

Fidelity Before Structure: Verbatim Chunks Beat Lossy Artifact Extraction in Long-Conversation LLM Memory

Tao An

Hawaii Pacific University
tan1@my.hpu.edu

Abstract

A growing class of conversational-memory systems compresses dialogue history into structured artifacts (extracted facts, decisions, or events) on the premise that distilled structure retrieves better than raw text. We test this premise with a controlled ablation: within one fixed retrieval–rerank–reasoning pipeline, we swap only the stored representation (LLM-extracted typed artifacts versus verbatim conversation chunks), holding the model, retriever, reranker, and judge constant. Verbatim chunks win by 15.9 points on LoCoMo (43.9% vs. 28.0%) and 22.0 points on LongMemEval-S (67.4% vs. 45.4%); a 1-hop semantic graph does not recover the gap, and six confound controls reproduce the effect. The mechanism is lossy distillation, not structure per se: accuracy tracks how much source text survives in the store, and the extracted-artifact pipeline does not beat naive RAG in overall accuracy (though chunks abstain worse, §6). For the extraction designs we test, structured memory should augment verbatim text rather than replace it: adding artifacts alongside chunks preserves accuracy; substituting them forfeits the gap. Code and data: <https://github.com/tao-hpu/cog-canvas>.

1 Introduction

“Please use type hints everywhere.”

This unremarkable instruction, dropped in turn 3 of a 50-turn programming discussion, is exactly the kind of detail that long-conversation memory systems must preserve, and routinely lose. Once the context window fills, truncation discards it outright; summarization paraphrases it to “the user prefers type hints,” quietly deleting the quantifier *everywhere*. In a controlled probe, summarization-based memory answers such constraint questions at 14.0% exact match, versus 91.0% when the verbatim wording is retrieved, a 77.0-point gap between **compression and fidelity**.

This fidelity problem has motivated a now-large family of memory systems that, rather than store raw dialogue, *distill* it into structured artifacts (extracted facts, decisions, events, or episodes) and retrieve over those (Packer et al., 2023; Chhikara et al., 2025; Zhou and Han, 2025; EverMind AI, 2025; Edge et al., 2024). They differ in mechanism but share one premise: that distilled structure is a *better* retrieval target than the raw text it replaces. The premise is explicit: Mem0 credits its gains to “capturing only the most salient facts in memory, rather than retrieving large chunk of original text” (Chhikara et al., 2025); SeCom reports compression “can effectively serve as a denoising mechanism, enhancing memory retrieval accuracy” (Pan et al., 2025). Yet it has not been cleanly isolated: published comparisons are system-level, swapping retrieval stacks, answerers, and judges along with the representation (Packer et al., 2023; Chhikara et al., 2025), while representation-level studies vary the granularity of stored *text* (Pan et al., 2025; Wu et al., 2025) or sweep structure types across task suites (Zeng et al., 2024) rather than pairing extraction against the text it replaces in one fixed pipeline. If extraction also *discards* information, structure and fidelity pull in opposite directions, and which one wins is an empirical question.

We answer it with a controlled ablation. Inside one fixed retrieval–rerank–reasoning pipeline, we change *only* the stored representation (LLM-extracted artifacts versus verbatim conversation chunks), holding the backbone model, retriever, reranker, and LLM judge constant. The result inverts the premise: verbatim chunks beat extracted artifacts by **15.9 points** on LoCoMo (Maharana et al., 2024) (43.9% vs. 28.0%) and **22.0 points** on LongMemEval-S (Wu et al., 2025) (67.4% vs. 45.4%), the ordering holds across every answerable question type, and the extracted-artifact pipeline does not outperform even a naive RAG baseline in overall accuracy ($p=0.89$). Most pointedly, within

this fixed pipeline extraction earns its cost in *neither* memory-budget regime: when the conversation fits the window (LoCoMo, $\sim 26\text{K}$ tokens) artifacts fall 31.9 points below the free full-context ceiling, and when it does not (LongMemEval-S, $\sim 115\text{K}$ tokens) chunk retrieval clears even the full-context baseline (67.4 vs. 60.6, different judge) while artifacts trail it. To be clear on scope: ours is a negative result about lossy *extraction*, not a claim that verbatim chunks are universally optimal (they abstain worse, §6), confined to the stateless, training-free regime where representations swap cleanly (§4).

The cause is *lossy distillation*. Extraction throws away verbatim detail that chunks keep for free, and a 1-hop semantic graph built over the artifacts does not recover the loss. Six confound controls—graph disabled, zero chunk overlap, evidence budget matched, dense-only retrieval, session-level extraction granularity, and a sentence-verbatim store that pins the loss to fidelity rather than granularity—reproduce the gap, and a 9-configuration GraphRAG sweep leaves its tuned ceiling far below verbatim chunks.

This *sharpens* rather than contradicts concurrent work reporting that a carefully designed extraction can match strong baselines (Zhou and Han, 2025): its units are minimally abstractive with source provenance, near the *verbatim* end of the spectrum. Both results follow one account: retrieval accuracy tracks how far the stored representation departs from the source. The closest prior sweep (Zeng et al., 2024) already ranks chunk stores strongest on dialogue; we add what a ranking cannot: paired single-pipeline isolation, a mechanism (fidelity, not structure), and the claim that *replacing* text—not adding structure—is what costs. We make the following contributions:

- **A controlled-ablation finding:** across two benchmarks and every answerable query type, verbatim chunks outperform *lossy* typed-artifact extraction by 15.9 and 22.0 points under an otherwise identical pipeline, and the artifact pipeline does not beat naive RAG overall ($p=0.89$; Tables 1 and 2).
- **A mechanistic account—lossy distillation—**that reconciles our negative result with concurrent positive findings on near-lossless extraction.
- **Evidence against the obvious objec-**

tions: six confound controls, a four-axis GraphRAG grid search (§O), a synthetic 77.0-point summarization-vs-verbatim probe, and—ruling out a strawman extractor—a stronger-extractor swap and decontaminated-prompt re-run that leave the gap intact (Appendices D, J.4).

- **A measured cost analysis** (Table 3): the verbatim-chunk pipeline costs more per query but less per *correct answer* (\$12.5 vs. \$14.9 per 1,000 correct answers), so extraction is not rescued on efficiency grounds either.

2 Related Work

Context window management approaches include sparse attention (Child et al., 2019; Beltagy et al., 2020) and retrieval-augmented methods (Borgeaud et al., 2022; Izacard et al., 2022); we focus on *what* to preserve rather than extending capacity.

Memory-augmented LLMs: MemGPT (Packer et al., 2023) pioneered OS-inspired virtual memory management and is now the production framework Letta (Letta AI, 2024). EMem (Zhou and Han, 2025) reaches strong accuracy with a training-free design whose event units are deliberately *non-compressive*—near-verbatim propositions with source-turn provenance (§4.7 and Appendix H reconcile its result with ours). Mem0 (Chhikara et al., 2025) incrementally extracts and updates salient facts (in its own harness the extracted memory, 66.9, trails its 72.9 full-context ceiling—the within-harness ordering we report); A-Mem (Xu et al., 2025) maintains an evolving note graph whose entries link and rewrite one another; SeCom (Pan et al., 2025) keeps verbatim segments but varies their granularity. We reproduce each system’s mechanism in our one fixed pipeline (§4.7) as points on a single fidelity curve, so accuracy attaches to *what is stored*, not confounded stacks. Closest to our question, Zeng et al. (2024) sweep chunks, triples, atomic facts, summaries, and mixed stores across six datasets, finding chunk-based and mixed stores strongest on dialogue understanding (LoCoMo)—consistent with our chunks and union results; Appendix Q positions our isolation, mechanism, and replacement claim against that ranking. We test the premise in the *training-free, stateless* drop-in regime most memory libraries ship by default.

Graph-based methods like GraphRAG (Edge

et al., 2024) excel at static documents but struggle with dynamic conversational state. **Summarization** (Tang et al., 2021; Chen et al., 2021) operates on lossy compression, systematically losing specific details. See Appendix Q for extended discussion.

3 Systems Under Test

Our goal is not to present a system but to isolate a design variable. We factor the conversational-memory design space into two axes—*what is stored* (verbatim chunks vs. LLM-extracted artifacts) and *whether a semantic graph links the stored items*—and instantiate all four combinations inside one pipeline; everything else is shared. All four occupy the same *stateless, training-free, infrastructure-free* cell (Table 6, Appendix A). The Artifacts+Graph cell is the full CogCanvas system as originally engineered; the chunk cells replace exactly one write-time stage of it (Figure 1).

3.1 Shared Backbone

At **write time**, the conversation is converted into a set of stored items according to the representation axis below; every item receives a **bge-m3** embedding. At **query time**, a hybrid retriever fuses dense similarity with lexical keyword matching; the top-30 coarse candidates are re-scored by **bge-reranker-v2-m3**; the top 15 re-ranked items form the context under a rarely-binding 5,000-token cap (the budget-matched row of Table 1 controls the resulting prompt asymmetry); and a reasoning-aware (chain-of-thought) prompt hands the selected items—with their turn indices and session timestamps—to the answerer (gpt-4o). All four pipelines execute this path with identical hyperparameters (§4).

3.2 Axis 1: Stored Representation

Verbatim chunks. The transcript, with explicit turn and session-timestamp markers, is segmented by a sliding window of 512 characters with 100-character overlap, aligned to line breaks; the parameters match the standalone RAG baseline exactly. Each chunk is stored as-is: no LLM at write time, no information discarded. Write-time confound controls remove the overlap (control ii of §4) and the graph (control i, Axis 2).

Extracted artifacts. Each turn is passed through an extraction LLM (gpt-4o-mini), which emits typed artifacts $o = (\tau, c, q, s, e, t, \gamma)$: type τ drawn

from an eight-type inventory—five task-oriented (DECISION, TODO, KEYFACT, REMINDER, INSIGHT) and three personal/social added for conversational data (PERSONATTRIBUTE, EVENT, RELATIONSHIP)—content string c , a *verbatim grounding quote* q excerpted from the source message, source role s , embedding e , turn index t , and extraction confidence γ . The design instantiates the standard premise of extraction-based memory: distill each turn into its retrievable essence, anchored by a quote to limit hallucination. A second *gleaning* pass (Guo et al., 2024) re-scans each turn for missed entities (pronoun referents, omitted subjects, implicit causal and temporal expressions); its measured contribution is negligible (Limitations, Appendix D). The store contains whatever the extractor deemed worth keeping—precisely the property under test.

3.3 Axis 2: Graph Expansion

Stored items of either representation are linked at write time by identical rules: *reference edges* when embedding cosine similarity exceeds $\theta_{\text{ref}}=0.5$ or lemmatized keyword overlap exceeds 0.3 (Jaccard), and *causal edges* for type- and time-compatible pairs (similarity above $\theta_{\text{causal}}=0.45$, predecessor within a 20-turn window), plus a temporal heuristic linking recent KEYFACTS and REMINDERS to subsequent DECISIONS. At query time, retrieved items are expanded one hop before reranking. Within the artifact pipeline this expansion lifts *retrieval* recall from 25.8% to 71.8% on LoCoMo10 (Table 14 in Appendix K)—which makes its null effect on final *answer* accuracy over chunks (Table 1) the more diagnostic.

4 Experiments

We evaluate the four pipelines, alongside external baselines, on a controlled information-retention benchmark.

4.1 Experimental Setup

Benchmark Design. We construct synthetic multi-turn conversations in which specific facts are “planted” at designated turns, each belonging to one of three categories drawn from the extraction ontology of §3 (KEYFACT, DECISION, REMINDER). Two variants: **Standard** ($N=50$; 50 turns, rolling compression at turn 40, single-hop recall) and **Multi-hop Reasoning** ($N=50$; questions requiring causal or impact-based reasoning across multiple planted facts, e.g. “Why was X decided?”).

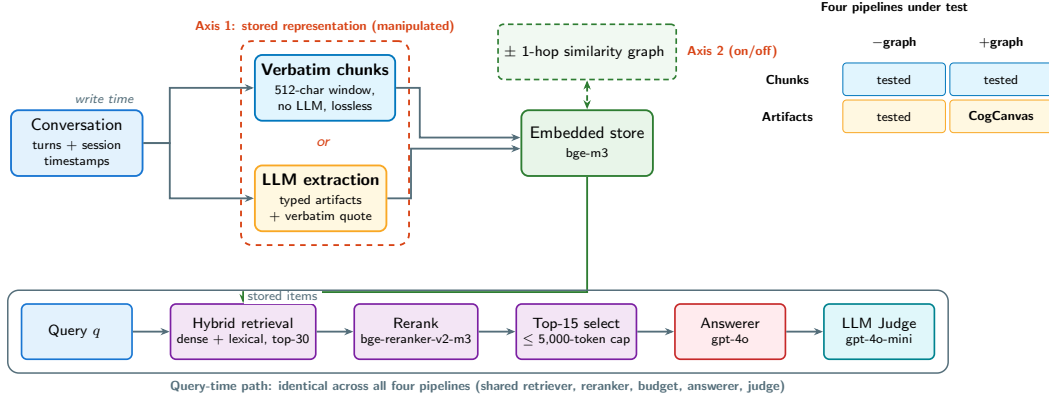


Figure 1: The controlled ablation design. At write time the conversation passes through exactly one manipulated stage—verbatim 512-character chunking *or* LLM extraction into typed artifacts (Axis 1)—into a shared embedded store, optionally linked by a 1-hop similarity graph (Axis 2). The query-time path (hybrid retrieval, reranking, token-budgeted context selection, answerer, judge) is identical across all four resulting pipelines; the Artifacts+Graph cell is the full CogCanvas system as originally engineered.

Querying happens after the compression boundary discards short-term context, isolating long-range recall from short-context lookup (walked example: Figure 3, Appendix A).

Evaluation Metrics. The standard benchmark reports **Recall Rate** (token-level fuzzy match ≥ 0.80) and **Exact Match**; the multi-hop benchmark reports **Keyword Coverage**, its **Pass Rate** ($KC \geq 0.80$), and KC on causal/impact questions (definitions in Appendix J). For LoCoMo and LongMemEval we report **LLM Judge accuracy**: binary CORRECT/INCORRECT by gpt-4o-mini given (q, g, a) at temperature 0 (prompt in Appendix J)—the headline metric because gold answers admit paraphrases that substring EM penalizes unfairly. Judge-human agreement on a stratified 100-question subset is 95% ($\kappa=0.897$), near-symmetric and unbiased across representations (Appendix J.6).

Reproducibility. All experiments share one configuration (recall-boost preset; full list in Appendix J): extractor gpt-4o-mini (temp. 0.1), answerer gpt-4o (LoCoMo, LongMemEval) or gpt-4o-mini (synthetic probe), shared by every system per benchmark (temp. 0.0), judge gpt-4o-mini (temp. 0), embedder bge-m3, reranker bge-reranker-v2-m3 (both local); final top- $k=15$ from a reranked pool of 30 under a rarely-binding 5,000-token cap; rolling compression every 40 turns retaining the 5 most-recent turns.

Baselines. Reflecting our *training-free* focus (heavily-optimized systems like EverMemOS are discussed in Related Work), we compare against: **Truncation** (most recent 5 turns); **Summarization** (history compressed by the answerer-tier model every 40 turns, recent 5 turns kept verbatim—MemGPT’s (Packer et al., 2023) core hierarchical mechanism); **RAG** (512-char chunks, 100-char overlap, top-10 dense retrieval); **RAG+Rerank** (same chunks, top-20 candidates re-scored by bge-reranker-v2-m3, the identical reranker CogCanvas employs—making it **our primary fair baseline**; it inherits RAG’s top-20→10 depth, the swap pipelines top-30→15); and **GraphRAG**, the official Microsoft implementation (Edge et al., 2024), reported at the best cell of a two-tier, 4-axis, 9-configuration grid search (tuned ceiling 13.0% on LoCoMo, drift mode; default 11.2%; Appendix O). Stateful systems (e.g. Letta, 74.0% (Letta AI, 2024)) sit outside this stateless comparison (Appendix C).

Fair-comparison disclosure. GraphRAG underwent a 9-configuration sweep and the summarizer prompt engineering; CogCanvas hyperparameters were fixed on the synthetic benchmark, and its extraction prompt’s LoCoMo-style few-shot examples favor the artifact pipeline—biasing *against* our finding, yet a decontaminated-prompt re-run leaves the result unchanged (28.5% vs. 28.0%, $p=0.82$; Appendix J.4). RAG and RAG+Rerank use deliberately standard settings; under-tuning them could only widen the artifact deficit (Appendices J.7, M).

4.2 Main Results (Standard)

Table 4 (Appendix A) shows the methods clustering by storage fidelity rather than architecture (Figure 4, Appendix A). Every verbatim-preserving store sits at or above 89% exact match—including the artifact pipeline (97.5 recall / 93.0 EM), whose grounding quotes make it lossless on this probe by design—while the lossy stores collapse (summarization 19.0 / 14.0, truncation 12.5 / 7.0). Each planted fact is a single, clean, extractable statement, so extraction gives up nothing here—a parity the real-conversation sections show does not survive natural data.

4.3 Multi-hop Reasoning Results

The artifact pipeline achieves **81.0%** pass rate (Table 5 and Figure 5, Appendix A), far ahead of RAG (55.5%) and the chunk pipelines (24.5–25.0%). This inversion is informative in both directions: it certifies the extraction machinery (a mis-engineered pipeline could not dominate its own ablation by 56 points, a cross-batch comparison bounded by the 1–3.5pp drift of Table 13, so the real-data losses ahead cannot be blamed on a broken extractor), and it delimits where the premise holds—every planted fact is authored as exactly one artifact and each question recombines two or three of them, a pre-shaping natural conversation never offers. The spread *within* the verbatim family (24.5–55.5 under shared chunking) shows retrieval-stack interactions dominate this probe; it is a fragile basis for architectural claims and we base none on it.

4.4 Evaluation on Real-World Dataset (LoCoMo)

We evaluate on **LoCoMo** (Maharana et al., 2024) (natural conversational noise, 300+ turns per conversation), reporting **LLM-Judge accuracy** (binary correctness scored by GPT-4o-mini, the de-facto standard of memory-system evaluations (Chhikara et al., 2025; Letta AI, 2024), validated against human annotation in Appendix J.6). Every system shares one retrieval–rerank backbone and one answerer (gpt-4o); only the stored representation changes.

Table 1 reports the representation swap (per-type breakdown: Figure 6, Appendix A). The verbatim-chunk pipeline reaches **43.9%** overall; replacing those chunks with LLM-extracted artifacts—the CogCanvas representation—drops accuracy

to **28.0%**, a 15.9-point loss inside an otherwise identical pipeline. The loss is uniform—multi-hop +13.1pp, temporal +19.0pp, open-domain +13.5pp: no question type favors extraction (the temporal mechanism—rewriting generalizes away surface date strings—is traced through failure cases in Appendix E). A closed-book control (same answerer, *no* retrieved context) scores just 2.3% ($n=699$), so accuracy reflects retrieved text, not parametric priors or contamination.

Crucially, the extracted-artifact pipeline (28.0%) does not outperform naive RAG (27.6%): extraction adds latency and token cost (§4.6) without buying accuracy. Six confound controls rule out retrieval, budget, and granularity artifacts. (i) *Graph disabled*: a 1-hop graph over chunks changes nothing (43.1% vs. 43.9%). (ii) *Zero lexical overlap*: forcing adjacent chunks to share no tokens preserves the advantage (42.8%), so chunks do not win by overlapping the gold answer. (iii) *Evidence budget matched*: artifacts are short (~ 30 –60 tokens) and chunks long (~ 130), so the shared $k=15$ hands chunks nearly twice the prompt (2,127 vs. 1,191 tokens); raising artifacts to $k=60$ under a 2,000-token cap recovers only 1.2pp (28.0%→29.2%; $p=0.36$), the gap still 14.7pp ($p<10^{-13}$; bootstrap 95% CI [12.0, 17.2]). (iv) *Dense-only retrieval*: dropping the lexical component leaves the gap at 14.3pp (41.8% vs. 27.5%). (v) *Session-level extraction*: extracting once per session instead of per turn scores 29.2% ($p=0.57$ vs. per-turn), gap 14.7pp—not an artifact of per-turn fragmentation. (vi) *Granularity vs. fidelity*: a sentence-verbatim store (source kept verbatim at artifact-scale units) beats artifacts by 5.7pp at matched granularity and budget, and budget-matching it leaves only 3.7pp below chunks, so granularity is minor and fidelity load-bearing (Appendix G). The deficit is informational—what the store contains—not allocational. Tuned GraphRAG (13.0%, Appendix O), Summarization (5.7%), and Truncation (5.0%) trail far behind.

Union storage: replacement, not coexistence, causes the loss. If the loss came from *replacing* text, a union store should recover chunk-level accuracy—and it does: indexing artifacts *alongside* the chunks yields 42.5%, indistinguishable from chunks alone (43.9%; McNemar $p=0.39$) and far above artifacts alone (28.0%; $p<10^{-12}$). The union’s artifacts are accuracy-neutral—structure may coexist with verbatim text, but substituting it

Stored representation	Overall	Multi-hop	Temporal	Open-domain
<i>Reference ceiling (no compression; conversations fit the context window)</i>				
Full context (entire conversation in prompt)	59.9	57.4	62.3	59.4
<i>Verbatim conversation chunks</i>				
Chunks, no graph	43.9	35.5	50.2	47.9
Chunks + 1-hop graph	43.1	36.5	47.7	46.9
Chunks, zero overlap (control ii)	42.8	34.8	49.2	44.8
Chunks, dense-only retrieval (control iv)	41.8	34.0	47.7	44.8
<i>Mixed store</i>				
Chunks $\cup$ artifacts + graph (union)	42.5	35.8	48.3	42.7
<i>LLM-extracted artifacts</i>				
Artifacts + graph (CogCanvas)	28.0	22.3	31.2	34.4
Artifacts + graph, budget-matched ($k=60$)	29.2	25.2	30.2	37.5
Artifacts + graph, dense-only (control iv)	27.5	21.6	30.5	34.4
<i>External baselines</i>				
RAG + Rerank	30.5	41.1	16.5	45.8
RAG ($k=10$)	27.6	34.8	14.0	52.1
GraphRAG (drift) [†]	13.0	12.8	7.5	32.3
Summarization (rolling)	5.7	5.3	0.9	22.9
Truncation (recent-5; no retrieval)	5.0	3.9	0.3	24.0
Δ Chunks – Artifacts	+15.9	+13.1	+19.0	+13.5

[†]GraphRAG uses gpt-4o-mini as the answerer (its indexing pipeline is coupled to the model); all other rows use gpt-4o. The row reports the best of a two-tier, 9-configuration grid over {chunk_size, max_gleanings, community_level, search_method} (the *drift* mode; Appendix O). Under a matched gpt-4o-mini answerer, Chunks-no-graph scores 41.9%—+28.9pp over GraphRAG—so the gap is not a model artifact.

Table 1: **LoCoMo (Cat 1–3; 10 conversations, 699 questions; LLM-Judge via GPT-4o-mini)**. Within one fixed retrieval–rerank–reasoning pipeline we swap only the stored representation; all rows share one retriever (bge-m3), reranker (bge-reranker-v2-m3), answerer (gpt-4o), and judge. The full-context row is a reference ceiling, not a memory system—LoCoMo conversations (~ 26 K tokens) fit gpt-4o’s window, so nothing is compressed; **bold** marks the best memory-system entry per column (baseline rows eligible). Verbatim chunks beat LLM-extracted artifacts by 15.9 points overall and on every question type; the gap is significant (McNemar exact test: 154 vs. 43 discordant pairs, $p < 10^{-15}$; cluster bootstrap resampling whole conversations: 95% CI [13.0, 18.4]pp), while artifacts vs. naive RAG is not (106 vs. 103, $p = 0.89$). The graph, overlap, budget, dense-only, and union controls are analyzed in the text.

forfeits 15.9 points.

The held-out categories widen the gap—except where abstention is the skill. The main table covers categories 1–3 (multi-hop, temporal, open-domain), which exercise long-range memory hardest; categories 4–5 follow here, so every LoCoMo type appears. On held-out *category 4* (single-hop factual recall; 841 questions) the gap *widens* to +33.3pp: chunks 82.6% vs. artifacts 49.3% (McNemar $p < 10^{-60}$)—answers that live in surface detail are exactly what extraction discards. *Category 5* (adversarial; 446 unanswerable questions, scored with a LongMemEval-style abstention template) is the one type that favors artifacts: 15.0% vs. chunks’ 6.5% (−8.5pp, $p = 1.9 \times 10^{-6}$). The mechanism mirrors the LongMemEval abstention column: the verbatim store surfaces the trap fact and the answerer asserts it, while the sparser artifact store more often returns nothing assertable—rewarding

storing less, not extracting better; both remain far from competent refusal.

Anchoring the absolute level. Published LoCoMo numbers run higher—Mem0 self-reports 66.9, with a 72.9 full-context ceiling (Chhikara et al., 2025)—but are computed over categories 1–4 with different answerers and judges. Re-computed over that category mix our same-batch chunks score 65.1% (1,002/1,540), inside the published range (artifacts 39.7%), and our full-context ceiling rises to 74.4%—broadly consistent with Mem0’s 72.9. Table 1’s lower headline reflects the harder 1–3 subset, not a weak harness; the ordering is a within-stack comparison, and extraction trails full context in Mem0’s harness just as in ours.

Stored representation <i>n</i>	IE 150	MSR 121	KU 72	TR 127	Abs 30	Overall 500	T-Avg —
Chunks, no graph	84.7	62.8	77.8	50.4	46.7	67.4	64.5
Chunks $\cup$ artifacts (union)	80.7	58.7	76.4	51.2	36.7	64.6	60.7
RAG + Rerank	85.3	61.2	72.2	38.6	70.0	64.8	65.5
RAG	86.7	58.7	72.2	37.0	70.0	64.2	64.9
Artifacts + graph (CogCanvas)	50.7	34.7	69.4	36.2	43.3	45.4	46.9
+ budget matched ($k=60$)	50.7	36.4	69.4	40.9	50.0	47.4	49.5
Truncation (recent-5)	13.3	0.0	13.9	0.8	93.3	11.8	24.3
Δ Chunks – Artifacts	+34.0	+28.1	+8.3 ^{ns}	+14.2	+3.3 ^{ns}	+22.0	+17.6

Table 2: **LongMemEval-S (500 questions; LLM-Judge via GPT-4o-mini)**. The same representation swap as Table 1, on a second benchmark. IE = information extraction, MSR = multi-session reasoning, KU = knowledge updates, TR = temporal reasoning, Abs = abstention; Overall is question-weighted, T-Avg the unweighted category mean. Truncation’s 93.3 abstention is degenerate—it answers almost nothing. McNemar exact tests: chunks vs. artifacts $p < 10^{-15}$ (152 vs. 42 discordant pairs; bootstrap 95% CI [17.0, 27.0]pp); artifacts trail even naive RAG, $p < 10^{-11}$ (48 vs. 142). Δ s are computed from unrounded counts; ^{ns} marks per-type deltas that are not significant (KU: 15 vs. 9 discordant, $p=0.31$; Abs: 4 vs. 3, $p=1.0$, a one-question margin).

4.5 Cross-Benchmark Validation (LongMemEval)

To test whether the finding generalizes beyond LoCoMo, we evaluate on **LongMemEval** (Wu et al., 2025) (500 questions, ~ 115 K tokens per conversation), which decomposes memory into five abilities: *Information Extraction* (IE), *Multi-Session Reasoning* (MSR), *Knowledge Updates* (KU), *Temporal Reasoning* (TR), and *Abstention* (Abs., correctly refusing unanswerable questions). We follow the benchmark’s official evaluation (GPT-4o-mini judge with question-type-specific prompts); as before, every system shares one retrieval–rerank backbone and answerer (gpt-4o), and only the stored representation changes. GraphRAG and Summarization are omitted: each question carries its own ~ 115 K-token haystack, so both would re-index or re-compress 500 times—prohibitive for baselines already trailing every retrieval method by 15–38 points on LoCoMo.

Table 2 reproduces the pattern on a second benchmark: chunks reach **67.4%** overall, artifacts **45.4%**—a 22.0-point loss—and chunks lead on all five question types, significantly on three: IE (+34.0pp, $p < 10^{-9}$), MSR (+28.1pp, $p < 10^{-5}$), and temporal (+14.2pp, $p=0.011$). Knowledge updates—where structured state-tracking was most expected to help—show a positive but non-significant lead (+8.3pp, $n=72$, $p=0.31$): even here extraction fails to win. The abstention lead (+3.3pp) is a one-question margin over $n=30$.

What loses is the representation, not the retrieval stack: both verbatim pipelines—chunks (67.4%) and naive RAG (64.2%)—sit ~ 20 points above artifacts, while the reranker moves accuracy by under a point. Both LoCoMo controls replicate: budget-matched $k=60$ artifacts recover 2.0pp (47.4%, gap 20.0pp, $p < 10^{-12}$), and a union store scores 64.6%—indistinguishable from chunks ($p=0.14$), +19.2pp over artifacts alone—so artifacts stay accuracy-neutral once verbatim text is present. The losses concentrate where surface detail matters most (IE 50.7 vs. 84.7, MSR 34.7 vs. 62.8)—the wording and per-turn provenance extraction normalizes away. As an external anchor, the benchmark authors report GPT-4o at 60.6% reading the full ~ 115 k-token haystack (Wu et al., 2025); chunk retrieval at 67.4% clears that full-context baseline (different judge), so artifacts lose to no weak representation.

The finding is about extraction, not about chunks being universally best. Chunks’ one clear weakness is abstention (46.7% vs. naive RAG’s 70.0% on LME’s 30 abstention questions, $p=0.039$; the weakness replicates with far more power on LoCoMo’s 446 adversarial questions, §4): handed no relevant evidence, the pipeline still answers rather than refuses (see Limitations). But on every answerable question type across both benchmarks, the extracted-artifact representation is dominated by at least one verbatim representation, and never significantly beats either overall—no tested carve-out where extraction wins (case study: Appendix B). The ordering holds cross-lingually: on native-Chinese PerLTQA both verbatim stores beat the typed-artifact store by ~ 47 –50 points (Appendix I).

4.6 Cost and Efficiency Analysis

We re-ran all instrumented methods on LoCoMo10 (categories 1–3) with per-call token logging, plus a full re-run with a gpt-4o-mini answerer (Table 3; USD at OpenAI list prices of 2026-05-27; embedding/reranking run locally at zero marginal cost). Three findings (Appendix L): extraction is *not* the dominant cost (\$0.14 of CogCanvas’s \$2.92 total—the gpt-4o answerer dominates under every architecture); it *does* buy cheaper generation ($5.1\times$ shorter completions), yet per correct answer the ranking inverts (\$12.5 per 1,000 for chunks vs. \$14.9 artifacts, \$16.4 RAG+Rerank); and every pipeline scales near-linearly to gpt-4o-mini (Total[†]). Extraction is not rescued on efficiency grounds.

Method	Extract	Gen	Total	Total [†]	Ratio
Truncation	\$0.00	\$1.25	\$1.25	\$0.09	13.2×
RAG	\$0.00	\$3.50	\$3.50	\$0.23	15.3×
RAG+Rerank	\$0.00	\$3.50	\$3.50	\$0.23	15.3×
Chunks (no graph) [§]	\$0.00	\$3.84	\$3.84	\$0.23	16.7×
Summarization	\$0.67	\$1.74	\$2.41	\$0.17	14.2×
Artifacts (CogCanvas)	\$0.14	\$2.78	\$2.92	\$0.31	9.5×
GraphRAG [‡]	—	—	—	—	—

Table 3: Measured cost on LoCoMo10 (699 questions, 2,938 indexed turns). **Extract/Gen** aggregate response.usage tokens per phase; **Total** adds embedding/reranking (\$0 on local GPUs); **Total[†]** re-runs end-to-end with a gpt-4o-mini answerer; **Ratio** = gpt-4o / mini. **Bold** marks the cheapest retrieval-augmented method at the gpt-4o tier. [§]Chunks measured on the 2026-06 batch; cross-batch comparability validated by RAG+Rerank, whose gen cost agrees across batches within 0.2%; its Total[†] is exact arithmetic (no extraction stage; re-pricing validation in Appendix L). [‡]GraphRAG’s LLM calls run inside the Microsoft graphrag subprocess, bypassing our logger (cost estimates: Appendix L).

4.7 Discussion

Extraction commits to relevance before the question exists. An extractor decides at *write time* what survives; whatever it discards is unrecoverable at *read time*. Verbatim chunks defer the decision to query time, when the question can guide selection. The hardest-hit categories are those whose answers live in surface detail (IE −34.0pp, MSR −28.1pp), and 69.0% of diagnosable chunk-pass/artifact-fail questions involve facts surviving in no deliberately extracted artifact—a temporal query such as “*When did Caroline join the LGBTQ support group?*” is answered from the dated source turn the chunk store keeps verbatim, but the artifact store answers only if the extractor logged that date (Appendix D)—errors *avoidable by not extracting*. The synthetic multi-hop probe shows the converse: when facts *are* pre-shaped to the schema, artifacts win (§4).

Neither graph expansion nor reranking is the missing ingredient. Graph expansion connects stored items but cannot regenerate content absent from all of them: it changes nothing over chunks (43.1% vs. 43.9%) and GraphRAG never exceeds 13.0% across its 9-configuration sweep—its community-summarization layer dissolves the per-turn timestamps conversational temporal queries depend on (Appendix O). Reranking is equally marginal: bge-reranker-v2-m3 moves overall accuracy by +0.6pp on LongMemEval and +2.9pp on

LoCoMo, an order of magnitude below the representation effect.

Fidelity, not structure, governs the ordering.

Ordering this paper’s designs by distance between stored representation and source text—verbatim chunks, near-verbatim event units with provenance (Zhou and Han, 2025), typed artifacts, community summaries—reproduces the observed accuracy ordering (the EDU point now measured: Appendix H), and isolating fidelity confirms causality: keeping a random fraction of each verbatim chunk’s tokens (pipeline fixed) drives accuracy down monotonically (retain 1.0→0.2: 43.9→35.5→23.9→14.6→9.0; each step McNemar $p < 10^{-5}$; Figure 2). Reproducing four memory systems’ defining mechanisms in the same pipeline places them on this axis in fidelity order—extraction stores (Mem0, A-Mem) fall ~22 points below chunks ($p < 10^{-26}$), while verbatim-preserving SeCom stays high (a within-tier segmentation effect); the *official* Mem0 package reproduces this ordering, ruling out a strawman (Appendix D). The design question is thus not *whether* to add structure but *where*: annotating verbatim text (e.g. Zep’s episode-grounded graph (Rasmussen et al., 2025)) preserves a lossless path to source; replacing it severs that path (Appendices C, E).

5 Conclusion

We reported a controlled negative result for *lossy* extraction-based memory: with only the stored representation swapped, LLM-distilled artifacts lose to verbatim chunks by 15.9 points on LoCoMo and 22.0 on LongMemEval-S, and the damage tracks fidelity, not granularity or graph tuning. Chunks’ own weakness is abstention (§6); the finding indicts *lossy extraction*, not text: structure should *augment* verbatim text, not replace it.

6 Limitations

Scope of the negative result. Our finding concerns extraction that *replaces* source text with distilled artifacts, tested for one artifact schema within one pipeline family. The schema’s extraction runs as an initial pass plus one gleaning pass (§3); an ablation found that removing the gleaning pass leaves LoCoMo accuracy statistically unchanged (Appendix D), so the second pass neither drives nor masks the gap, and we did not tune it per-dataset. Extractor strength is unlikely to be the

missing ingredient either: swapping the gpt-4o-mini extractor for gpt-4o leaves both artifact accuracy and the chunk advantage statistically unchanged (Appendix D). The strongest candidate for an escape—near-verbatim, provenance-preserving event units (Zhou and Han, 2025)—we did test: reproduced inside the fixed pipeline, the EDU store lands between typed artifacts and chunks ($30.2 < 36.3 - 36.5 < 47.4$, both gaps significant, saturated in k ; Appendix H), confirming the fidelity ordering while still trailing raw text; its published full-system numbers additionally rely on LLM-filtered, graph-propagated retrieval that our swap holds fixed. We cannot rule out that some untested design escapes the pattern.

Extraction’s one win does not generalize. The one setting where artifacts win—our synthetic multi-hop probe (Table 5, 81.0 vs. 55.5)—is by construction extraction’s best case: every planted fact is pre-shaped into exactly one artifact. The inversion does not transfer to *human-authored* multi-hop, where verbatim chunks still lead on LoCoMo’s multi-hop category (35.5 vs. 22.3; $N=282$, +13.1pp, cluster-bootstrap 95% CI [10.2, 15.9], McNemar $p=2.2 \times 10^{-5}$) and on LongMemEval multi-session reasoning (62.8 vs. 34.7; $N=121$, +28.1pp, 95% CI [18.2, 38.0], $p < 10^{-6}$). The artifact advantage therefore does not generalize beyond ontology-aligned data; the two human-authored multi-hop cells are individually significant, not a small-sample artifact.

What the controls do and do not rule out. Our controls rule out rescue by the most commonly proposed remedies: (i) graph expansion; (ii) chunk-overlap artifacts; (iii) evidence-budget asymmetry (the budget-matched control recovers 1.2pp of the 15.9pp gap); (iv) lexical-retrieval bias toward source text (the dense-only control leaves a 14.3pp gap); (v) per-turn fragmentation (the session-granularity control leaves a 14.7pp gap); (vi) coarse-vs-fine granularity (a budget-matched sentence-verbatim store trails chunks by only 3.7pp; Appendix G); (vii) baseline under-tuning (Appendix O); and (viii) cost arguments (§4.6). Within the schema we tested two extraction granularities—per-turn and per-session—with statistically indistinguishable results (§4); segment-level construction (Pan et al., 2025) and incremental update schemes (Chhikara et al., 2025) remain untested in our harness.

Benchmarks and language. Our two headline benchmarks are English chat-style QA (LoCoMo,

10 conversations / 699 questions in the main protocol plus 1,287 held-out category-4/5 questions; LongMemEval-S, 500 questions); a native-Chinese probe (PerLTQA, Appendix I) reproduces the ordering more starkly still, though on model-generated dialogues. Memory uses beyond QA—proactive recall, personalization, summarization-style review—remain untested.

One retrieval stack; the answerer is not load-bearing. All pipelines share one embedding and reranker family (bge-m3, bge-reranker-v2-m3) and one answerer (gpt-4o). The ordering is not specific to that retriever: re-running the chunks-vs-artifacts comparison with the answerer fixed at gpt-4o-mini under a held-out dense embedder (OpenAI text-embedding-3-small) and a sparse lexical retriever (BM25) leaves the chunk advantage essentially unchanged—chunks score 43.1–43.8% and artifacts 28.5–29.9% across all three retriever families, a gap of +13.6 to +14.7pp (Appendix N). The answerer, however, is not what drives the result: swapping gpt-4o for the much weaker gpt-4o-mini costs verbatim chunks only 2.0pp (43.9% to 41.9%), and re-running the controlled fidelity dial (§4.7) under the gpt-4o-mini answerer reproduces the same monotone collapse (retain 1.0→0.2: 43.5→30.8→21.2→12.2→8.4; Figure 2)—so the fidelity→accuracy ordering is not an artifact of the strong model. Nor is it an artifact of *random* token deletion: sweeping SeCom’s own extractive compression reproduces the same monotone collapse (1.0→0.5: 47.4→36.9, falling below fixed chunks under aggressive compression; Appendix F), so the ordering holds when fidelity is removed by a real extractor rather than at random. We nonetheless keep every headline comparison same-model for cleanliness; the GraphRAG row is the one exception (its indexing is coupled to gpt-4o-mini) and is bounded by the matched-model control in Table 1.

Run-to-run variance. Every table entry is a single run at temperature 0; paired McNemar tests and cluster-bootstrap CIs capture question- and conversation-level sampling uncertainty. To bound API nondeterminism directly, we re-ran both headline LoCoMo cells three times end-to-end (fresh extraction caches for the artifact arm): within-batch spread is negligible—chunks 47.4/47.4/47.5, artifacts 30.2/30.3/30.6, paired gap +17.2/+17.0/+16.9pp. Across months the unpinned gpt-4o endpoint drifts: both arms shift upward in parallel against the 2026-06 headline batch (43.9/28.0), while re-judging the stored

June answers with the same judge tier reproduces the published numbers within 0.2pp on LoCoMo (44.1/28.2) and 0.8pp on LongMemEval (67.4/46.2 vs. 67.4/45.4)—the drift lives in the answerer endpoint, not the judge or protocol, and the representation gap is, if anything, larger under the newer snapshot (17.0 vs. 15.9pp). Appendix J.1 consolidates every batch assignment.

The winning representation has a real weakness. Verbatim chunks trail naive RAG on LongMemEval abstention by 23.3pp (46.7% vs. 70.0%; $n=30$, McNemar exact $p=0.039$), and the better-powered evidence is LoCoMo’s 446 adversarial questions, where chunks reach only 6.5% against the artifact pipeline’s 15.0% ($p=1.9\times 10^{-6}$): handed plausible but irrelevant evidence, the chunk pipeline answers when it should refuse. The result is a negative finding about extraction, not a claim that chunks are universally optimal. Relatedly, the category-5 abstention judge is validated against an independent lexical detector (92.4–97.5% agreement), a context-isolated blinded LLM annotator, and an independent human annotator (the latter two both 50/50, $\kappa=1.0$; Appendix J.6). Bounding loss under forced compression remains future work, and *repairing* abstention is harder than it looks: adding an evidence-sufficiency gate that refuses when the top reranker score falls below a threshold τ does lift category-5 refusal from 6.5% toward naive-RAG levels (to 55.6–65.5% as τ rises from 0.1 to 0.4), but answerable accuracy on categories 1–3 collapses in lockstep (43.1% down to 25.3–17.9%), dropping below even the artifact pipeline—no threshold buys abstention without sacrificing answerable questions roughly one-for-one. Two stronger *semantic* refusers—an LLM evidence-sufficiency gate and an answer-then-verify check—fare no better: on LongMemEval-S they can drive abstention from 53% past naive RAG’s 70% (to 93%), but only at a –13pp answerable-accuracy cost, and on LoCoMo they merely match or modestly improve the score-gate frontier (Appendix N). Competent refusal is therefore a genuine property the verbatim representation lacks under three distinct repair mechanisms, not a tuning oversight; the refusal rate *can* be lifted past naive RAG, just not for free—pointing to the same remedy as our main finding, a refusal layer that augments rather than replaces the verbatim store.

Training-free scope. Stateful and fine-tuned systems (e.g., Letta at a self-reported 74.0%, Ev-

erMemOS at 92.3%) sit outside our controlled comparison; their numbers come from different answerer models and evaluation stacks and are not commensurable with Table 1 (Appendix C). We therefore treat every such figure strictly as an *external anchor*, not a baseline in our tables, and make no head-to-head claim against it; only the within-pipeline rows of Tables 1–2 (one shared retriever, reranker, answerer, and judge) constitute a controlled representation comparison. Notably, the strongest of these external systems are themselves extraction- or classification-based. A controlled probe indicates the ordering persists under *statefulness*: inside a stateful agent (a MemGPT-style core-memory-plus-archival loop with incremental compression), swapping *only* the archival representation from verbatim turns to extracted artifacts—holding the stateful machinery fixed—costs 10.9 points on LoCoMo (24.0% vs. 13.2%; McNemar exact 100 vs. 24 discordant, $p<10^{-11}$), reproducing the direction of our stateless headline on every question type (Appendix C). Whether it also survives fine-tuning remains untested.

7 Ethics Statement

This work evaluates memory systems on two public benchmarks of fictional, synthetically constructed conversations (LoCoMo, LongMemEval); no real user data, personally identifying information, or human subjects are involved. The judge-validation study (Appendix J.6) was annotated by an author and one independent volunteer annotator; both labeled only model outputs on public benchmarks (no personal or human-subjects data). The volunteer participated with informed consent, received no monetary compensation (contributing as an acknowledged collaborator, to be credited in the camera-ready Acknowledgments, omitted here for anonymity); because the task involves only public model outputs and no human-subjects data, it is exempt from IRB review. All model access used commercial APIs under their terms of service, and we report the full token and dollar cost of the experiments (§4.6) to make the computational footprint transparent. The paper’s central contribution is a cautionary, negative result about extraction-based memory rather than a new capability, and we foresee no dual-use risk beyond those already present in retrieval-augmented LLM systems.

Scientific artifacts and licensing. All evaluation data are public research benchmarks, used

consistently with their intended non-commercial research use: LoCoMo (Maharana et al., 2024) and PerLTQA (Du et al., 2024) are released under CC BY-NC 4.0, and LongMemEval (Wu et al., 2025) under the MIT license. We redistribute none of these datasets; our released run artifacts contain only model outputs and scores. Our independently constructed synthetic planted-fact benchmark, the experiment harness, and run artifacts *accompany this submission* as anonymized supplementary material and are available to reviewers now; only their formal open-source license (MIT for code, CC BY 4.0 for data) is deferred to publication. The supplementary archive contains the pipeline and experiment-harness code, the synthetic-benchmark generator, all extraction/answerer/judge prompts (including the decontaminated-prompt variant), cached model outputs, scoring and significance-testing scripts, and per-run manifests sufficient to reproduce every table and figure.

Use of AI assistance. We used an AI programming assistant for software scaffolding and routine refactoring of the experiment harness, and an LLM for light copy-editing (grammar and concision) of author-written prose. All research questions, experimental design, analysis, and claims are the authors’ own; no text was generated wholesale by an LLM, and the authors verified every reported number against the run artifacts.

References

- Iz Beltagy, Matthew E Peters, and Arman Cohan. 2020. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*.
- Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, and 1 others. 2022. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*, pages 2206–2240. PMLR.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. 2023. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*.
- Yulong Chen, Yang Liu, Liang Chen, and Yue Zhang. 2021. DialogSum: A real-life scenario dialogue summarization dataset. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 5062–5074.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshray Yadav. 2025. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*.
- Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. 2019. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*.
- Yiming Du, Hongru Wang, Zhengyi Zhao, Bin Liang, Baojun Wang, Wanjuan Zhong, Zezhong Wang, and Kam-Fai Wong. 2024. PerLTQA: A personal long-term memory dataset for memory classification, retrieval, and synthesis in question answering. In *Proceedings of the 10th SIGHAN Workshop on Chinese Language Processing*.
- Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.
- EverMind AI. 2025. EverMemOS: A memory operating system for AI agents. <https://github.com/EverMind-AI/EverMemOS>.
- Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. 2024. LightRAG: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779*.
- Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. 2022. Atlas: Few-shot learning with retrieval augmented language models. *arXiv preprint arXiv:2208.03299*.
- Kuang-Huei Lee, Xinyun Chen, Hiroki Furuta, John Canny, and Ian Fischer. 2024. A human-inspired reading agent with gist memory of very long contexts. In *International Conference on Machine Learning*.
- Letta AI. 2024. Benchmarking AI agent memory: Is a filesystem all you need? <https://www.letta.com/blog/benchmarking-ai-agent-memory>. Accessed: 2025-01-04.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. ArXiv:2402.17753.
- Charles Packer, Vivian Fang, Shishir G Patil, Kevin Lin, Sarah Wooders, and Joseph E Gonzalez. 2023. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*.
- Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Xufang Luo, Hao Cheng, Dongsheng Li, Yuqing Yang, Chin-Yew Lin, H. Vicky Zhao, Lili Qiu, and Jianfeng Gao. 2025. On memory construction and retrieval for personalized conversational agents. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. 2025. [Zep: A temporal knowledge graph architecture for agent memory](#). Preprint, arXiv:2501.13956.

Xiangru Tang, Arjun Nair, Borui Wang, Bingxiang Wang, Jai Desai, Aaron Wang, Ilmi Haque, Shafiq Joty, Dragomir Radev, and 1 others. 2021. ConvoSumm: Conversation summarization benchmark and improved abstractive summarization with argument mining. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3622–3634.

Weizhi Wang, Li Dong, Hao Cheng, Xiaodong Liu, Xifeng Yan, Jianfeng Gao, and Furu Wei. 2023. Augmenting language models with long-term memory. In *Advances in Neural Information Processing Systems*, volume 36.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2025. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *Proceedings of the International Conference on Learning Representations (ICLR)*. ArXiv:2410.10813.

Jeff Wu, Long Ouyang, Daniel M Ziegler, Nissan Stiennon, Ryan Lowe, Jan Leike, and Paul Christiano. 2021. Recursively summarizing books with human feedback. *arXiv preprint arXiv:2109.10862*.

Wujiang Xu, Kai Mei, Hang Gao, Juntao Tan, Zujie Liang, and Yongfeng Zhang. 2025. A-Mem: Agentic memory for LLM agents. *arXiv preprint arXiv:2502.12110*.

Michihiro Yasunaga, Hongyu Ren, Antoine Bosselut, Percy Liang, and Jure Leskovec. 2021. QA-GNN: Reasoning with language models and knowledge graphs for question answering. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 535–546.

Ruihong Zeng, Jinyuan Fang, Siwei Liu, and Zaiqiao Meng. 2024. [On the structural memory of LLM agents](#). Preprint, arXiv:2412.15266.

Xikun Zhang, Antoine Bosselut, Michihiro Yasunaga, Hongyu Ren, Percy Liang, Christopher D Manning, and Jure Leskovec. 2022. GreaseLM: Graph reasoning enhanced language models. In *International Conference on Learning Representations*.

Wanjun Zhong, Lianghong Guo, Qiqi Gao, and Yanlin Wang. 2024. MemoryBank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19724–19731.

Sizhe Zhou and Jiawei Han. 2025. [A simple yet strong baseline for long-term conversational memory of LLM agents](#). Preprint, arXiv:2511.17208.

Method	Recall	Exact	Cmplx.
<i>Lossy stores</i>			
Truncation	12.5%	7.0%	Low
Summarization	19.0%	14.0%	Low
GraphRAG	83.5%	70.0%	High
<i>Verbatim stores</i>			
RAG (k=10)	93.5%	89.5%	Low
Chunks, no graph	92.0%	90.5%	Low
Chunks + graph	92.5%	91.0%	Low
Artifacts + graph (CogCanvas)	97.5%	93.0%	Low

Table 4: Information retention on the standard synthetic benchmark (a fixed 50-conversation subset, 50 turns, 4 planted facts each; rolling compression every 40 turns; gpt-4o-mini answerer). The two chunks rows were measured in the 2026-06 batch; all other rows are the 2025-12 batch under the identical protocol (Table 13; same-configuration cross-batch drift on this harness is 1–3.5pp with direction preserved, §6). The summarization-vs-chunks exact-match gap (14.0% vs. 91.0%), the 77.0-point figure quoted in §1, is therefore a cross-batch difference, far larger than any observed drift. Cmplx. = qualitative implementation complexity (write-time machinery and infrastructure). Truncation keeps the most recent 5 turns (§4).

A Additional Figures and Tables

This appendix collects the design-space positioning table and the visualizations of the main results referenced from the body of the paper.

B Case Study: Verbatim Preservation

The exact-match gap on the synthetic probe illustrates the fidelity mechanism at the scale of a single fact. Consider a REMINDER planted in turn 3: “**use type hints everywhere.**” When queried about coding style preferences after compression, the two lossy representations respond differently:

Summarization: “The user prefers a consistent code style with type hints.” *Exact: ✗*

Artifact pipeline: “please use type hints everywhere” *Exact: ✓*

The summarizer correctly identifies that type hints are relevant but loses the critical qualifier “everywhere”; the artifact’s verbatim citation field preserves the exact constraint. This is the regime where extraction is at its best: each planted fact is a single self-contained statement that maps onto exactly one artifact, so nothing depends on the extractor’s judgment of salience. The case therefore certifies the extraction machinery rather than the extraction premise—a verbatim chunk store preserves the

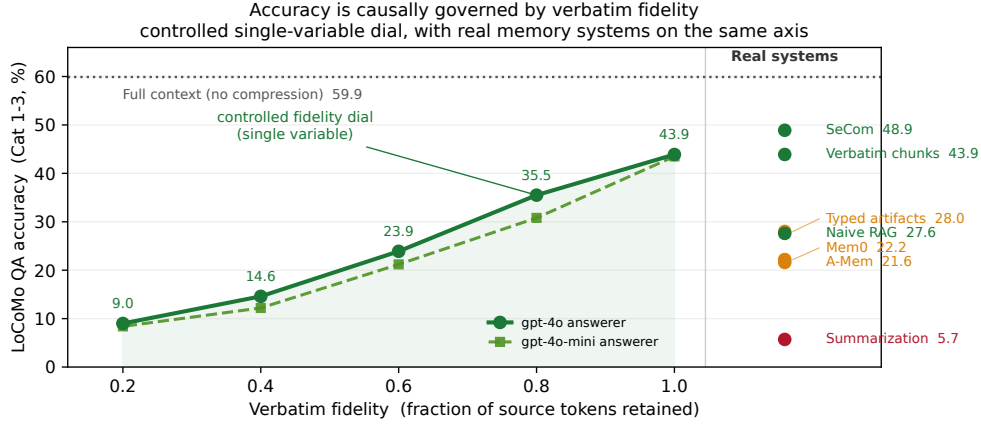


Figure 2: **A controlled fidelity dial** (§4.7). Keeping a random fraction of each verbatim chunk’s tokens (solid line; pipeline held fixed) drives LoCoMo Cat 1–3 accuracy down monotonically, and retain=1.0 reproduces the verbatim-chunks anchor (43.9%). The dashed line repeats the dial under a gpt-4o-mini answerer (retain 1.0→0.2: 43.5→30.8→21.2→12.2→8.4): the near-coincident curve shows the monotone collapse is a property of the stored representation, not of the strong answerer model. Real memory systems (right column) land on the same accuracy axis in fidelity order—extraction-based Mem0/A-Mem in the dial’s low-fidelity regime, verbatim-preserving SeCom/chunks in the high—and full context (no compression) sits above all. SeCom’s edge over fixed chunks is a segmentation effect *within* the verbatim tier, not a fidelity reversal.

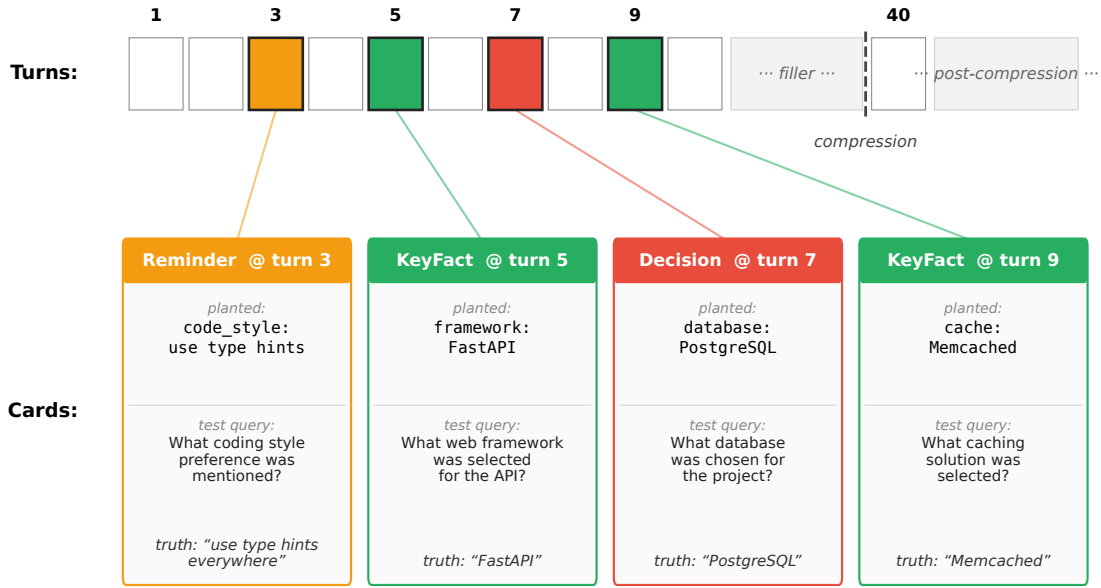


Figure 3: Walked example of the synthetic benchmark protocol on conv 922f651b (50 turns, “hard” difficulty). *Turns row:* a 50-turn conversation with four facts planted at turns 3, 5, 7, 9 (colored cells); filler dialogue (turns 11–39) and the post-compression query window (turns 41–50) are collapsed into ellipsis blocks for space. The dashed line at turn 40 marks the rolling-compression boundary that discards short-term context. *Cards row:* each planted fact carries a cognitive type drawn from the broader extraction ontology of §3—here REMINDER (orange), KEYFACT (green), or DECISION (red)—together with its structured key: value content, the post-compression test query, and the ground-truth answer used to score Recall and Exact Match.

same wording *unconditionally*, without requiring the extractor to notice the fact at all. The real-data

gap of Table 1 opens precisely where that condition fails: 69.0% of the artifact pipeline’s diagnosable

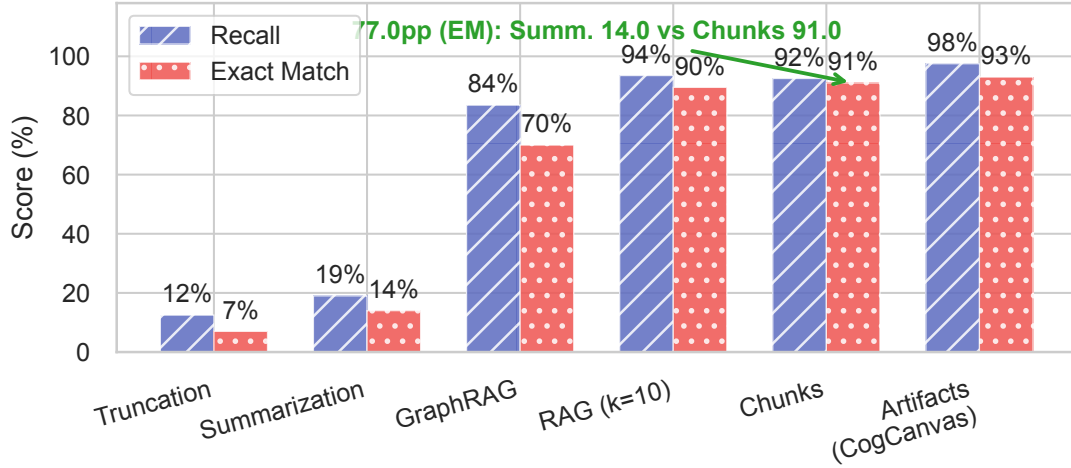


Figure 4: Planted-fact retention on the synthetic benchmark (protocol of Table 4). Methods cluster by storage fidelity: verbatim stores (RAG, chunks, quote-grounded artifacts) all exceed 90% recall (89.5–93.0% exact match), lossy stores collapse. The annotated 77.0-point exact-match gap is the figure quoted in §1.

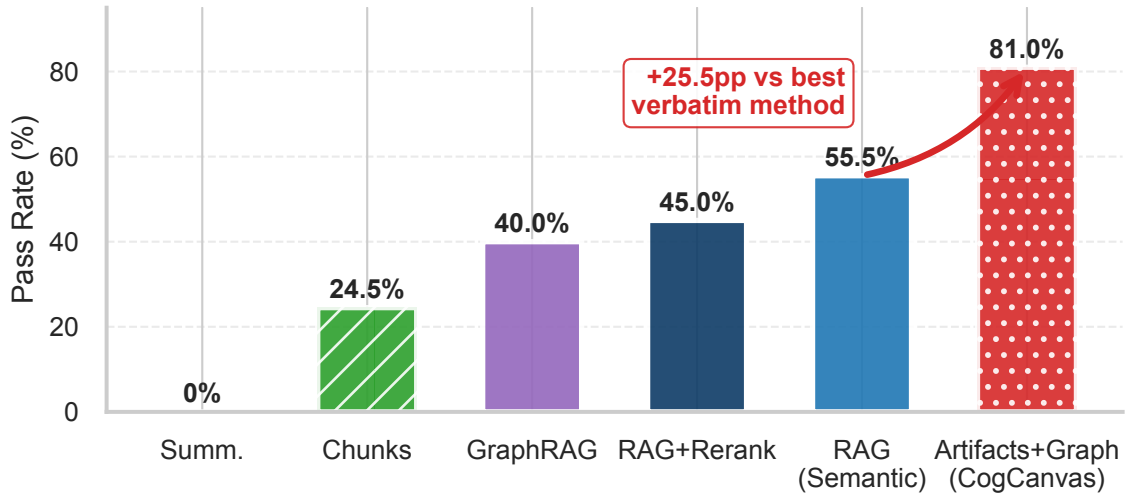


Figure 5: The synthetic multi-hop probe is the one setting where the artifact pipeline wins (81.0% vs. 55.5% for the best verbatim method)—extraction’s home turf, where every fact is pre-shaped into exactly one artifact. The advantage inverts on both real benchmarks.

discordant failures involve facts the extractor never deliberately wrote down (Appendix D).

C Stateful vs. Stateless Memory: A Trade-off Analysis

Our results reveal a fundamental trade-off between **stateful** and **stateless** memory architectures for long conversations:

Stateful Systems (e.g., Letta, EverMemOS). These methods maintain persistent memory state across interactions, typically backed by databases or specialized storage. Letta achieves 74.0% on

LoCoMo (Letta AI, 2024), while EverMemOS reaches 92.3% (EverMind AI, 2025). Their advantages include:

- **Superior recall:** Persistent storage enables exhaustive memory retention without context length constraints.
- **Incremental learning:** Agents can accumulate knowledge over extended sessions.

However, these benefits come at significant cost:

- **Infrastructure overhead:** Requires databases (PostgreSQL), containerization (Docker), and

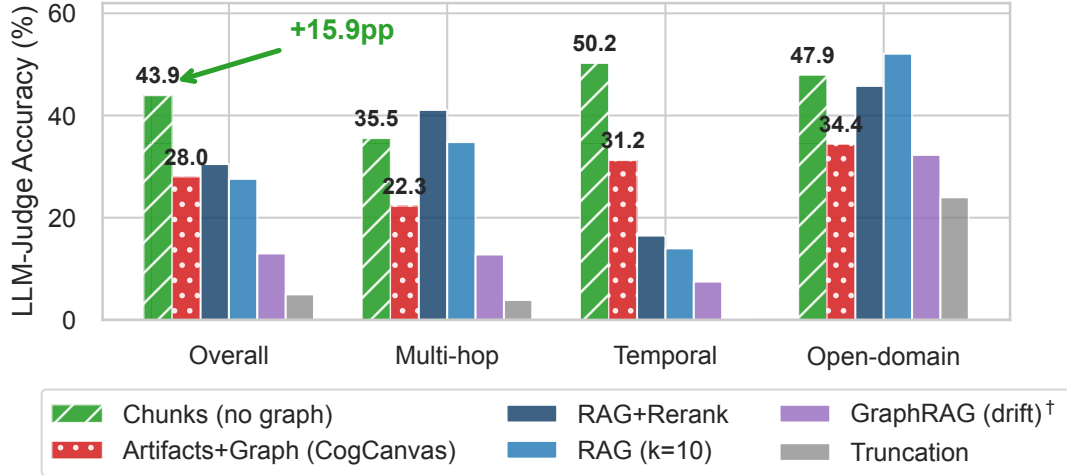


Figure 6: LoCoMo breakdown by question type (per-question micro-averages, LLM Judge). Swapping LLM-extracted artifacts (red) for verbatim chunks (green) inside the same pipeline costs accuracy on every question type. [†]GraphRAG uses a gpt-4o-mini answerer (Table 1, footnote).

Method	Pass	KW	Causal	Impact
Summarization	0.0%	42.5%	48.3%	35.8%
Chunks + graph	24.5%	60.8%	50.6%	71.0%
Chunks, no graph	25.0%	60.8%	50.6%	71.0%
GraphRAG	40.0%	64.8%	62.5%	67.0%
RAG+Rerank	45.0%	68.2%	64.2%	73.3%
RAG	55.5%	74.8%	68.8%	79.5%
Artifacts + graph (CogCanvas)	81.0%	90.2%	87.5%	92.6%
Δ Artifacts – RAG	+25.5pp	+15.4pp	+18.7pp	+13.1pp

Table 5: Multi-hop causal reasoning on the synthetic probe (50 conversations, rolling compression every 40 turns, gpt-4o-mini answerer). The two chunks rows and the RAG+Rerank row are the 2026-06 batch; all other rows, including both terms of the Δ row, are the 2025-12 batch under the identical protocol (Table 13; the RAG configuration re-measures at 52.5% in 2026-06, Appendix M). KW = Keyword Coverage. The one setting in the paper where the artifact pipeline wins—extraction’s home turf (see text); the advantage inverts on the real benchmarks (Tables 1 and 2).

server management.

- **Training requirements:** Top performers like EverMemOS rely on learned compression models.
- **Deployment complexity:** Non-trivial setup prevents rapid prototyping and integration into existing systems.

Stateless Systems (this work). All four pipelines under test are stateless: the conversation is indexed once, queries are answered independently, and no persistent agent state is maintained. The regime’s practical appeal is real—zero infrastructure (drop-

in via standard LLM APIs), no fine-tuning, full portability—and within it, our controlled comparison identifies the best stored representation as verbatim chunks (43.9% on LoCoMo, 67.4% on LongMemEval-S; Tables 1 and 2), not extracted artifacts.

When to Use Each Approach. Our findings suggest:

- **Stateful systems:** Production deployments with long-lived agents, where infrastructure investment is justified by sustained interactions.
- **Stateless verbatim pipelines:** Rapid prototyping, research experiments, or applications requiring portability and simplicity. Within the stateless budget, our results indicate the spend belongs in retrieval over verbatim text, not in extraction machinery.

The gap between the best stateless pipeline (43.9%) and self-reported stateful results (Letta 74.0%, EverMemOS 92.3%) is large, but it is not a controlled comparison: those figures come from different answerer models and evaluation stacks. What our controlled data show is narrower and firmer: whichever regime one chooses, rewriting conversation text into distilled artifacts is dominated by keeping the text.

A controlled probe inside a stateful agent (referenced from §6). The comparisons above swap

Method	TF	State	Infra	P&P	Incr.
Truncation	✓	✗	✗	✓	✓
Summarization	✓	✗	✗	✓	✓
RAG	✓	✗	✗	✓	✓
RAG+Rerank	✓	✗	✗	✓	✓
GraphRAG	✓	✗	✗	✓	✗
Letta (Letta AI, 2024)	✓	✓	✓	✗	✓
EverMemOS (EverMind AI, 2025)	✗	✓	✓	✗	✓
EMem (Zhou and Han, 2025)	✓	✗	✗	✓	✓
Pipelines under test (§3)	✓	✗	✗	✓	✓

Table 6: Design-space positioning across five operational axes. **TF**: training-free (no fine-tuning required). **State**: maintains memory state across queries. **Infra**: requires persistent infrastructure (e.g. PostgreSQL, Docker). **P&P**: plug-and-play (drop-in for any answerer LLM, no API contract). **Incr.**: supports incremental updates without re-indexing the full conversation. All four pipelines under test (chunks/artifacts $\times$ graph on/off) occupy the same *stateless*, *training-free*, *infrastructure-free* cell, so our comparison swaps representations within one corner of this space rather than promoting any one system. GraphRAG’s lack of incremental support stems from its community-summarization stage, which must re-cluster the entity graph when new turns arrive. Letta’s stateful design places it outside our stateless evaluation pipeline (§4, Appendix C); we cite its 74.0% LoCoMo result as an upper bound rather than a direct comparison.

representations in a *stateless* pipeline. To test whether statefulness overturns the ordering, we ran the same one-variable swap *inside* a stateful agent: a MemGPT-style controller that keeps the most recent turns in a fixed-size core memory and, on rolling compression, moves older turns into an embedded, keyword-indexed *archival* store retrieved at query time. Holding the entire stateful loop fixed (core-memory size, compression cadence, archival top- k , retrieval, answerer, judge), we swap only the archival representation: *verbatim* archived turns versus LLM-*extracted* artifacts (the extractor of §3; ~ 1.7 artifacts/turn, tens to hundreds per conversation, so the store is well populated). On LoCoMo (Cat 1–3, 699 questions), verbatim archival beats artifact archival on *every* question type (Table 7): +10.9pp overall (McNemar exact, 100 vs. 24 discordant pairs, $p < 10^{-11}$), the same direction and temporal-heavy shape as the stateless headline. The absolute level is lower than our full stateless pipeline (this lite controller is a weaker retrieval stack), but the *representation* ordering is unchanged—adding state does not rescue lossy extraction.

Archival representation	Overall	Multi-hop	Temporal	Open-domain
Verbatim turns	24.0	27.7	13.4	49.0
Extracted artifacts	13.2	14.5	7.2	29.2
Δ (verbatim – artifacts)	+10.9	+13.2	+6.2	+19.8

Table 7: **Statefulness does not overturn the ordering.** Swapping only the archival representation inside a stateful MemGPT-style agent (core memory + rolling compression + archival retrieval), everything else held fixed, on LoCoMo Cat 1–3 (699 questions, gpt-4o answerer, gpt-4o-mini judge). Verbatim archival dominates extracted-artifact archival on every question type; overall +10.9pp, McNemar exact $p < 10^{-11}$ (100 vs. 24 discordant). The Δ row is computed from unrounded counts (168 vs. 92 of 699 correct); cells are rounded to one decimal.

D Error Analysis: Where the Artifact Pipeline’s Answers Go Missing

Analyzing 3,121 missing keywords across the artifact pipeline’s failed LoCoMo questions reveals three primary failure modes: (1) **Extraction gaps** (78.8%): the LLM extractor fails to identify certain facts as salient (e.g., “Caroline is single” not extracted as a `key_fact`); (2) **Temporal blindness** (14.9%): dates mentioned in passing (“last Tuesday,” “May 2023”) are not preserved as structured temporal attributes; (3) **Reasoning chain breaks** (6.4%): multi-hop questions requiring inference across multiple artifacts fail when intermediate links are missing. This typology is assigned by inspecting answers, so it cannot by itself separate *write-time* losses (the store never contained the fact) from *query-time* misses (the store contained it but retrieval or generation failed). The store-level probe below makes that separation directly.

Store-Level Survival Probe. For each of the 147 questions that the chunk pipeline answers correctly and the budget-matched artifact pipeline misses (the discordant pairs of §4), we test whether the gold answer’s content tokens co-occur within any *single* stored artifact (its content, quote, and context fields), taking co-occurrence within a single conversation turn as the fairness ceiling; golds that never co-occur verbatim even in one turn are analyzed separately, and two golds with no content tokens at all (e.g. “Yes”) are excluded as undiagnosable. Among the 58 questions whose gold appears verbatim in a single turn, 17 (29.3%) survive in *no* artifact at all, and another 23 (39.7%) survive *only* inside the raw-text window that temporal extraction copies verbatim as context—never

as a deliberately extracted fact. “Caroline is single” is one of these: the answer token appears in no `key_fact`, only mid-window, and it survives at all precisely because that field keeps text verbatim. In total, 69.0% of these diagnosable failures trace to facts the extractor never deliberately wrote down (write-time losses); the remaining 31.0% are query-time misses on facts that were extracted but not surfaced. The other 87 questions have golds that appear verbatim in no single turn—typically dates that must be resolved from a relative expression plus a session header (“yesterday” + “8 May, 2023”)—so the chunk pipeline wins them not by containing the answer string but by handing the answerer the inference-enabling surface context that rewriting strips; in 25.3% of them the artifact store’s best single-object coverage falls below even the single-turn ceiling. A verbatim-chunk store cannot commit the write-time class at all—its failures are confined to query-time retrieval misses and over-answering (abstention)—which is why the Discussion characterizes these as errors avoidable by not extracting.

What Gets Missed? Qualitative analysis of extraction failures reveals systematic patterns in what the LLM extractor overlooks:

- **Implicit personal attributes** (e.g., “Caroline is single,” relationship status inferred from context rather than stated directly)—the extractor prioritizes explicit statements over implicit facts.
- **Emotional preferences** (e.g., “she felt anxious about...”)—affective states are often not recognized as KEYFACTS worth preserving.
- **Casual temporal references** (“last Tuesday,” “a few months ago”)—relative time expressions are harder to anchor than absolute dates.
- **Negations and constraints** (“does NOT want X”)—negative preferences are sometimes missed or inverted during extraction.

These patterns suggest that extraction prompts could be improved by explicitly instructing the LLM to attend to implicit attributes, emotional states, and negative constraints.

Gleaning contribution. The gleaning pass (§3) is designed to address these gaps by explicitly targeting omitted subjects and implicit relationships in a second extraction scan, and the main LoCoMo and LongMemEval configurations run the initial extraction plus one gleaning pass. Its measured contribution is negligible: in an ablation under an earlier evaluation protocol (January batch, conversation-

mean scoring), disabling gleaning *raised* LoCoMo accuracy from 28.6% to 30.7%—a 2.1pp difference within conversation-level noise—so the extraction gaps documented above persist with or without the second pass. We did not tune the number of gleaning passes per-dataset.

Stronger-extractor swap. A natural objection is that the extractor itself—gpt-4o-mini in the main configuration—is too weak, and a stronger model would close the gap. It does not. Re-running extraction with gpt-4o on a 3-conversation LoCoMo subset ($n=185$ questions) leaves artifact accuracy statistically unchanged (30.3% with gpt-4o vs. 29.7% with gpt-4o-mini on the same subset; McNemar $p \approx 1$), and the verbatim-chunk advantage stays intact at -17.8pp on this subset ($p < 10^{-4}$). Extractor strength is therefore not the missing ingredient: the loss is in what extraction *discards* relative to the source, not in how well a given model performs the extraction.

External-system anchor: the official Mem0 package. A second objection targets the in-pipeline reproductions of §4.7: by re-implementing each system’s defining mechanism inside our fixed pipeline, we might misrepresent the real system—a strawman. We rule this out for Mem0 (Chhikara et al., 2025) by running its *official* package end-to-end on LoCoMo (Cat 1–3, all 10 conversations, $n=699$): Mem0’s own extraction loop (`add(infer=True)`), its own vector search, and its native OpenAI embedder (`text-embedding-3-small`, cosine retrieval). Three stores share one harness—(1) official Mem0 extracted memory, (2) verbatim 512-character chunks over the same embedder, (3) the full transcript (reference ceiling)—with a *shared* gpt-4o-mini answerer and the *same* LLM judge used for the headline tables; both memory stores receive per-item session timestamps, and Mem0 is granted twice the retrieval budget of chunks (top-30 vs. top-15). Even so, official Mem0 trails verbatim chunks at every aggregate: **Mem0 36.6% < chunks 47.9%** (-11.3pp) and **Mem0 < full context 40.5%**. The deficit concentrates exactly where the fidelity account predicts—temporal questions, 27.1% vs. 51.1% (-24.0pp): Mem0’s extraction rewrites or drops the surface dates the chunk store keeps verbatim (we observed it record “June 27, 2026” for an event the transcript dates to the prior day). Mem0’s own LoCoMo report shows the same extracted-vs-full-context direction ($66.9 < 72.9$).

Strong-answerer re-test (the strawman closed

at full strength). The harness above answers with gpt-4o-mini; a residual worry is that a weak answerer depresses every extraction system below its deployed level (Mem0’s 36.6% trails its self-reported 66.9), so the ordering might be an answerer artifact. We therefore re-ran the identical three-store anchor with the *strong* answerer of our headline tables (gpt-4o), letting Mem0 use gpt-4o for its *own* extraction loop as well, over the wider Cat 1–4 split (all 10 conversations, $n=1,540$). Official Mem0 climbs to **54.7%**—far above its stripped-harness 36.6%, so the harness does not cripple it—yet verbatim chunks still reach **69.9%**, a **+15.2pp** margin (McNemar exact: 343 vs. 109 discordant pairs, $p=3.0\times 10^{-29}$) that holds in *every* category: multi-hop +2.2 (46.5 vs. 44.3), temporal +39.9 (59.2 vs. 19.3), open-domain +6.2 (60.4 vs. 54.2), and single-hop +11.2 (82.9 vs. 71.7). The temporal gap is the widest, exactly as the fidelity account predicts: Mem0’s write-time rewriting discards the surface dates a verbatim store keeps for free (§E). Full context for this split (74.4%; with no retrieval it is harness-independent given the answerer) sits above both, so the strong answerer restores the expected fidelity ordering—full 74.4 > chunks 69.9 > Mem0 54.7—and dissolves the chunks-above-full inversion the gpt-4o-mini answerer produces (a lost-in-the-middle effect of handing $\sim 26K$ tokens to a small model, not a property of the stores). At its own full strength, then, a deployed extraction system still loses to verbatim chunks by the same ~ 15 -point margin as our headline, within one controlled comparison.

Two further controls localize the cause to the stored representation rather than our reproduction. (i) *Not a strawman extractor*. Running our *own* typed-artifact extractor in this same stripped harness scores 42.5%—above official Mem0 (36.6%)—so the headline artifact deficit is not the artifact of an extractor weaker than a deployed system; our extractor in fact retains more (~ 230 facts/conversation vs. Mem0’s ~ 20). (ii) *Not a reranker artifact*. With the bge reranker removed and both stores matched at top-15 under symmetric timestamps, verbatim chunks still beat our artifacts by +4.6pp (47.1% vs. 42.5%); the full rerank pipeline of Table 1 amplifies this to the +15.9pp headline. The verbatim advantage is therefore direction-robust across pipelines—present without the reranker, widened by it—and the lossy-extraction deficit is a property of the real system, not of our reproduction.

The high proportion of temporal markers (14.9%) directly explains why temporal questions remain hard for the artifact pipeline (31.2% in the same-batch protocol of Table 1) even though its date attributes and quote fields lead the timestamp-less baselines: what extraction notices it preserves well, and what it misses is lost at write time (Appendix E). Better extraction prompts—explicit date/time targeting, temporal attribute modeling—could shrink this class, but cannot reach the unconditional retention of a verbatim store.

Performance by question type. In the same-batch protocol of Table 1, the artifact pipeline reaches 22.3% on multi-hop, 31.2% on temporal, and 34.4% on open-domain questions. It leads RAG (14.0%) and RAG+Rerank (16.5%) on temporal reasoning—the category where its date attributes and quote fields matter most—but trails RAG+Rerank badly on multi-hop (22.3% vs. 41.1%): the category pre-joined artifacts are supposed to serve is instead won by retrieval over verbatim text, echoing the finding of §4.7 that extraction’s synthetic multi-hop advantage does not transfer. The chunk pipeline leads the artifact pipeline in every category.

E Temporal Questions as a Fidelity Probe

Temporal questions separate the stored representations more sharply than any other category, and the ordering tracks a single write-time quantity: how much temporal source information—session timestamps and the surface strings that carry them—survives in the store. In the same-batch protocol of Table 1, timestamp-marked verbatim chunks reach **50.2%**; extracted artifacts, whose schema captures dates as structured attributes alongside verbatim quotes, reach 31.2%; the standalone chunk baselines—verbatim text, but without the pipeline’s explicit session-timestamp markers or adaptive retrieval depth—trail far behind (RAG+Rerank 16.5%, RAG 14.0%); and representations that pass through summarization collapse (GraphRAG 7.5%, Truncation 0.3%). The artifact pipeline’s 14.7pp lead over RAG+Rerank (which shares the same BGE reranker; +13.1pp under the decontaminated-prompt control of Appendix J.4) and its 19.0pp deficit against verbatim chunks are two halves of one account: extraction preserves the timestamps the extractor notices, a timestamp-marked verbatim store preserves all of them, and a summarization layer preserves almost none. This section traces

that mechanism through failure cases and component ablations.

Failure Cases: GraphRAG vs. the Artifact Pipeline. A representative instance of summarization-style compression losing specific dates: “*When did Caroline go to the LGBTQ support group?*” (ground truth: May 7, 2023). The verbatim-chunk pipeline retrieves the original turn and returns “May 7, 2023”; GraphRAG, having summarized the relevant turns into a community-level description, returns “the specific date of her attendance is not provided”—the event survives summarization, but its timestamp does not. A second illustrative case from the LoCoMo benchmark exhibits the same failure mode under a different surface form:

- **Example: When did Melanie paint a sunrise?**
 - **GraphRAG:** offered only a vague temporal reference, stating, “the specific date... is not provided... mentioned that she created a painting... last year.” The exact date was generalized away during community summarization.
 - **Artifact pipeline:** returned the exact date, “8 May 2022”—the extractor captured this event, and the artifact’s verbatim quote field carried the date string through to retrieval. A verbatim chunk store preserves the same string with no extraction step at all.

The contrast places the three representations on the fidelity spectrum of §4.7 rather than crowning a winner: community summarization discards the date; extraction keeps it *when the extractor notices it*—the 14.9% temporal-blindness class in Appendix D consists precisely of the cases it does not—and verbatim chunks keep it unconditionally.

Ablation Study: Component Contributions. *Scope note.* This ablation is internal to the *artifact* pipeline: it measures how much each component contributes *given* that one has already committed to extracted artifacts as the stored representation, and it predates the chunks comparison (all rows are from the January batch, conversation-mean scoring—the protocol of the gleaning ablation in Limitations; the full configuration measures 28.0% in the same-batch protocol of Table 1, consistent with the 28.6% here). Its findings therefore do

Configuration	All	M-hop	Temp.	O-dom.
Full System	28.6%	20.9%	30.5%	44.8%
– Graph Expansion	25.8%	19.5%	29.0%	33.3%
– Reranking	20.9%	16.7%	23.1%	26.0%
Minimal (Graph only)	14.3%	11.7%	13.7%	24.0%

Table 8: Ablation study on LoCoMo (Cat 1/2/3, all 10 conversations, LLM Judge; January batch, conversation-mean scoring). Full system achieves 28.6%. Reranking is the largest contributor (−7.7pp when removed), followed by graph expansion (−2.8pp overall, −11.5pp on open-domain).

not contradict the main result: graph expansion recovers some of what extraction loses *relative to artifacts without a graph*, yet the completed system still trails verbatim chunks by 15.9 points—structure mitigates the damage of extraction without repairing it.

To quantify the individual contributions of the artifact pipeline’s components, we conduct a systematic ablation study by removing each component from the full system and measuring the performance drop.

Key findings:

- **Reranking is the largest contributor.** Removing BGE reranking causes a −7.7pp drop (28.6% → 20.9%), making it the most impactful component within the artifact pipeline. The same reranker added to vanilla RAG yields 30.5% overall in the same-batch protocol (Table 1), so reranking is a general-purpose improvement rather than an artifact-specific one. On temporal questions, RAG+Rerank reaches only 16.5% against the artifact pipeline’s 31.2% (and the chunk pipeline’s 50.2%): what drives temporal accuracy is how much source wording—especially timestamps—survives in the store, consistent with the fidelity ordering of §4.7.
- **Graph expansion helps—but only within the artifact representation.** Removing graph expansion reduces overall accuracy by −2.8pp (28.6% → 25.8%), with the largest impact on open-domain questions (−11.5pp, 44.8% → 33.3%). This is a within-representation effect: the graph recovers part of what extraction loses relative to artifacts without a graph, but the same 1-hop graph over verbatim chunks is a no-op (43.1% vs. 43.9%, Table 1)—structure cannot regenerate content

the store never kept (§4.7).

- **Combined system effects.** The full system (28.6%) improves +14.3pp over the minimal baseline (14.3%), demonstrating strong synergy between components.
- **Temporal accuracy is carried by preserved wording, not edges.** The artifact pipeline’s temporal lead over the timestamp-less baselines (31.2% vs. RAG’s 14.0%, Table 1) comes from the schema’s date attributes and verbatim quote fields—the least lossy parts of the representation—rather than from temporal heuristic edges. The same quantity bounds what the pipeline cannot do: dates the extractor fails to notice (Appendix D) are unrecoverable at query time, which is why timestamp-marked chunks, which keep every date unconditionally, lead by a further 19.0pp.

Retrieval Quality Analysis. To verify that graph expansion improves *retrieval quality* rather than merely leveraging LLM reasoning capabilities, we conduct a retrieval-only evaluation that measures whether the retrieved context contains ground-truth keywords—**without invoking the LLM for answer generation**. As shown in Table 14, graph expansion increases retrieval recall from 25.8% to 71.8% (+45.9pp). This confirms that the performance gains in Table 8 stem from *improved context retrieval*, not from LLM answering ability. The graph structure enables retrieving related artifacts that would otherwise be missed by pure semantic similarity matching.

Why Baselines Fall Short on Temporal Reasoning. The consistently low performance of baselines on temporal questions (RAG: 14.0%, GraphRAG: 7.5%, Summarization: 0.9%; Table 1) can be attributed to several factors:

- **RAG (Chunk-based):** Standard RAG systems typically chunk conversations or documents into fixed-size segments. Temporal information, such as specific dates or sequences of events, can easily be fragmented across chunk boundaries, making it difficult for retrieval to reconstruct the full temporal context. The lack of explicit temporal indexing or relational understanding means that chunks are retrieved based on semantic similarity, which often overlooks the critical temporal dimension.

- **GraphRAG (Community Summarization):**

While GraphRAG constructs a sophisticated knowledge graph, its reliance on community detection and summarization for higher-level queries often leads to the generalization of information. Specific dates, times, or precise event sequences are frequently summarized away or absorbed into broader thematic summaries, losing the granular temporal detail necessary for answering precise temporal questions. The emphasis on conceptual relationships can inadvertently obscure the temporal ordering of events.

- **Summarization:** Directly compressing conversation history into summaries fundamentally operates on a lossy principle. Temporal details are often considered less “important” than core facts or decisions by the summarization LLM, leading to their omission or vague representation. Iterative summarization, in particular, exacerbates this issue, as temporal markers are progressively lost with each compression pass, leading to recursive information decay for time-sensitive data.

All three failure modes are instances of the same write-time quantity—how much temporal source information survives in the store—and they place each system on the fidelity spectrum of §4.7. Artifact extraction sits in the middle of that spectrum: its structured date attributes and verbatim quotes preserve the temporal markers the extractor notices, hence its lead over the timestamp-less and summarization-based baselines; everything it fails to notice is lost before any query arrives, hence its 19.0pp deficit against timestamp-marked verbatim chunks (Table 1). The category-level lesson matches the paper’s headline result: *preserving* temporal wording matters more than *connecting* it through graph structure—and keeping all of it beats extracting some of it.

F A Real-Extraction Fidelity Dial: SeCom’s Compression Sweep

The controlled dial of §4.7 degrades fidelity by deleting *random* tokens, inviting the objection that it measures the damage of random corruption rather than the selective loss of a real extractor. We replicate the dial with a real extractive-compression mechanism—SeCom’s LLMlingua-2 denoiser—sweeping its

token-retention rate over $\{1.0, 0.75, 0.5\}$ with the pipeline otherwise fixed (backbone-routed cogcanvas-secom-nograph, gpt-4o answerer, LoCoMo Cat 1–3, $n=10$, 699 questions; micro-accuracy from per-question LLM-Judge decisions).

compress_rate	Overall	M-hop	Temporal	O-dom.
1.00 (verbatim segments)	47.4	37	56	47
0.75 (SeCom default)	46.9	39	53	49
0.50 (aggressive)	36.9	30	41	44
ref: verbatim chunks	45.2			
ref: full context	59.9			

Table 9: SeCom accuracy as a function of its own extractive-compression rate (LoCoMo Cat 1–3, micro-accuracy). Accuracy falls monotonically as less verbatim text is retained, and at aggressive compression a real extractive compressor drops *below* the fixed-chunk baseline. *Ref.* rows re-measure the chunks and full-context anchors in this batch: chunks re-measures at 45.2 against the 43.9 headline of Table 1, a same-configuration run-to-run difference of 1.3pp (Appendix J.1).

Accuracy falls monotonically as SeCom retains less verbatim text; at aggressive compression (0.5) it drops to 36.9%—below the fixed-chunk baseline (45.2%)—placing a real extractive compressor squarely in the dial’s low-fidelity regime. The 0.75 default (46.9%) sits 2.0pp below the SeCom mechanism-reproduction anchor of Figure 2 (48.9%)—a same-configuration repeat, and the widest one we observed (disclosed in Table 13)—so the sweep is on the same scale. Two further observations: (i) the curve *plateaus* between 0.75 and 1.0 (+0.5pp, within noise)—mild denoising is information-preserving, while keeping *more* verbatim text buys nothing, so SeCom saturates near 47%, still 12 points below full context; (ii) at full retention SeCom’s topical segmentation edges fixed-window chunks by ~ 2 pp (47.4 vs. 45.2), the segmentation effect noted in §4.7. The monotone collapse under a *real* extractor—not random deletion—is the load-bearing point: the fidelity→accuracy ordering is a property of how much source text survives, however it is removed.

G Separating Fidelity from Granularity: A Sentence-Verbatim Control

Verbatim chunks (~ 512 -char windows) differ from typed artifacts on two axes at once—granularity (a coarse window vs. a short item) and fidelity (raw source vs. lossy distillation)—so the chunks-vs-artifacts gap could in principle be a granularity effect. We isolate the two with a *sentence-verbatim*

Store (all same backbone)	Tok/q	Acc.
Verbatim chunks (coarse)	2,127	42.8
Sentence-verbatim, budget-matched	4,674	39.1
Sentence-verbatim, $k=15$	1,010	32.2
Typed artifacts (lossy, fine)	1,159	26.5

Table 10: Sentence-verbatim control (question-micro accuracy over 699 questions; one gpt-4o-mini batch for all four rows). “Tok/q” is mean answerer-prompt tokens per question. At matched fine granularity and budget, verbatim beats lossy by 5.7pp (32.2 vs. 26.5); budget-matching the sentence store recovers it to within 3.7pp of chunks, bounding pure granularity. This batch’s chunks and artifacts anchors (42.8, 26.5) re-measure the mini-answerer anchors reported elsewhere (41.9, §6; 28.5–29.9, Appendix N) within ~ 1 –2pp run-to-run noise (Appendix J.1).

store: the transcript kept fully verbatim but cut to artifact-scale units (one or more whole sentences packed to ≤ 160 characters; no extraction, no compression, sentences never split), routed through the same nograph backbone as every other anchor (gpt-4o-mini answerer, LoCoMo Cat 1–3, 699 questions, question-micro accuracy, one batch for all four rows; the ordering is answerer-invariant, §6). Table 10 reports the decomposition.

Three readings isolate the axes. (1) *Fidelity, controlling for granularity and budget.* At the same fine granularity and a slightly *smaller* answerer budget (1,010 vs. 1,159 tokens), the sentence-verbatim store beats artifacts by 5.7pp (32.2% vs. 26.5%): keeping the source verbatim helps even when unit size matches the artifacts’. (2) *Granularity, controlling for fidelity and budget.* Raising the sentence store’s k until it feeds at least the chunks-level token budget recovers it to 39.1%, within 3.7pp of verbatim chunks—so coarse-vs-fine granularity accounts for at most 3.7pp of the 16.3pp gap. (3) *Budget-responsiveness as a fidelity signature.* The added budget lifts the *verbatim* sentence store by 6.9pp (32.2→39.1) but lifts the *lossy* artifact store by only 1.2pp (the $k=60$ control, §4): retrieval budget converts to accuracy only when the source survives in the store. Fidelity, not granularity, is the load-bearing variable; pure granularity contributes roughly a fifth of the gap (3.7 of 16.3pp), and that 3.7pp is an upper bound (the sentence store is over-budgeted here).

Store	k	Tok/q	Overall	Single	Temp.	Multi
Verbatim chunks	15	2,127	47.4	37.9	53.9	53.1
EMem-style EDU	90	$\sim 2,700^e$	36.5	23.4	46.7	40.6
EMem-style EDU	60	$\sim 1,900^e$	36.3	22.0	47.4	41.7
EMem-style EDU	15	$\sim 700^e$	29.8	17.0	40.5	31.2
Typed artifacts	15	1,159	30.2	26.2	30.8	39.6

Table 11: EMem-style EDU control (LoCoMo Cat 1–3, question-micro accuracy; one 2026-07 batch, gpt-4o answerer—the chunks and artifacts rows are the same-batch re-runs of §6, so no cross-snapshot comparison is involved). ^eEDU token budgets estimated from mean unit length (103 chars $\approx$ 26 tokens) times k plus prompt overhead; the chunks and artifacts figures are instrumented means (Appendix G).

H A Near-Verbatim EDU Control: Measuring the High-Fidelity End of Extraction

Concurrent work reports that a training-free memory built from “enriched elementary discourse units” (EDUs)—self-contained statements with normalized entities and source-turn attributions, deliberately constructed to preserve information “in a non-compressive form”—reaches strong LoCoMo accuracy (Zhou and Han, 2025). Our fidelity account predicts rather than contradicts this (§4.7); here we test the prediction by measuring the EDU *representation* inside the fixed pipeline—the same mechanism-reproduction treatment given to Mem0, A-Mem, and SeCom. A gpt-4o-mini extractor decomposes each utterance into EDUs under three constraints matching the published design: full coverage (no salience filtering), near-verbatim wording (only pronoun-to-name normalization plus minimal glue words), and the same dated turn marker every other anchor prints. The store is dense (~ 21 k units over LoCoMo10, ~ 7 per turn, against ~ 1.7 typed artifacts); a parse failure falls back to sentence-verbatim units, so no turn is silently dropped (the decomposition prompt ships with the supplementary code). Because EDUs are fine-grained, we sweep retrieval depth: $k=15$ matches the headline protocol, $k=60$ approximately matches the chunks token budget, $k=90$ overshoots it.

Three results. (1) *The fidelity ordering is confirmed at a third, measured point.* EDUs at or above the chunks token budget score 36.3–36.5% ($k=60/90$), significantly above typed artifacts (30.2%; at $k=90$: McNemar 113 vs. 69 discordant, $p=0.0014$) and significantly below verbatim chunks (47.4%; 122 vs. 46, $p<10^{-8}$)—the monotone ordering of §4.7, previously cited at its middle point, now measured there. The temporal column

Store (same backbone, Chinese)	Acc.
Naive RAG (verbatim)	86.1
Verbatim chunks	82.2
Typed artifacts (lossy)	35.6

Table 12: PerLTQA dialogue-category QA (gpt-4o-mini answerer, 31 conversations, 902 questions). Both verbatim stores beat the lossy artifact store by ~ 47 –50 points; artifacts again never beat naive RAG.

shows the mechanism directly: near-verbatim units keep the date strings the typed artifacts lose (46.7–47.4% vs. 30.8%). (2) *The residual gap is not budget.* Raising k from 60 to 90 moves accuracy by +0.1pp (one question of 699; 25 vs. 24 discordant, $p=1.0$): the EDU store saturates ~ 11 pp below chunks, so what separates it from raw text is the representation, not evidence quantity. (3) *The headline claim sharpens.* Even a deliberately non-compressive, provenance-preserving extraction does not recover verbatim-chunk accuracy under a fixed retrieval stack; EMem’s published full-system results additionally employ LLM-filtered retrieval with optional graph propagation (Zhou and Han, 2025)—machinery our controlled swap deliberately freezes—and a different answerer and judge. Within the stateless regime we test, the design lesson is unchanged: annotating verbatim text is safe; replacing it, however gently, costs accuracy.

I Cross-Lingual Generalization: Native Chinese (PerLTQA)

Both headline benchmarks are English. We test whether the verbatim-over-lossy ordering survives in another language on PerLTQA (Du et al., 2024), a native-Chinese personal long-term-memory QA dataset (SIGHAN 2024). We use only its *dialogue-category* questions—those answerable from raw conversation rather than from PerLTQA’s pre-structured profile/event/relationship memory, which would favor extraction by construction—giving 31 character conversations (median 102 turns) with third-party-annotated gold answers (capped at 30 questions/conversation, 902 total). The same nograph backbone runs unchanged (gpt-4o-mini builder and answerer; the ordering is answerer-invariant, §6).

The ordering reproduces, more starkly than in English (Table 12): both verbatim stores—naive RAG (86.1%) and chunks (82.2%)—beat the typed-artifact store (35.6%) by ~ 47 –50 points, and artifacts again never beat naive RAG. The gap exceeds

LoCoMo’s 15.9pp because PerLTQA’s dialogue questions demand specific recall (what a character was confused about; which device a child wore) that the abstractive artifacts discard. Inspecting the store confirms the deficit is lossy distillation, not an extractor that cannot read Chinese: the English-prompt extractor produces well-formed Chinese typed artifacts (PERSON_ATTRIBUTE, EVENT, RELATIONSHIP)—they simply commit to facts other than the ones queried. Two caveats keep this a generalization probe rather than a third headline benchmark: PerLTQA’s dialogues are model-generated (its gold QA is human-annotated), and naive RAG’s 3.9pp edge over chunks is a within-verbatim-tier segmentation effect (both retain the source), not a fidelity reversal.

J Reproducibility Details

J.1 Protocol and Batch Provenance

Numbers in this paper come from evaluation batches run between December and July; every cross-batch caveat is flagged where the number appears, and Table 13 consolidates the provenance in one place. All headline comparisons (Tables 1, 2, and the category-4/5 paragraphs) are same-batch, same-protocol; January- and December-batch numbers appear only in appendix ablations and are never compared against main-table rows.

J.2 Clean-Room Supplement Verification

To confirm the supplementary package is self-contained, we extracted the anonymized archive into an empty directory and installed it into a fresh Python 3.10 environment following only the bundled README. Installation completes without error and all declared dependencies resolve. We then re-ran every *offline* analysis script—those that consume no API and read only the included result JSONs. All reproduce their reported quantities: the judge–human agreement study regenerates 95.0% agreement and $\kappa = 0.897$ (§J.6); the amortized-cost tables re-price from the stored token logs; and an independent re-scorer that recomputes LoCoMo numbers directly from stored per-question answer/judgment pairs (bypassing any cached score field) recovers the headline ordering on the bundled run files—verbatim chunks in the low-40s versus extracted artifacts at 28.6% (a January artifact run included in the package)—matching the direction and magnitude of Table 1, whose 2026-06 headline batch scores artifacts at

Result	Batch	Aggregation	Answerer
Tab. 4 (synthetic std.)	2025-12/06 ^d	per-fact	gpt-4o-mini
Tab. 5 (multi-hop)	2025-12/06 ^d	conv. macro	gpt-4o-mini
Tab. 1 (LoCoMo)	2026-06	question micro	gpt-4o ^a
cat. 4/5 paragraphs	2026-06	question micro	gpt-4o
Tab. 2 (LME-S)	2026-06	question micro	gpt-4o
Tab. 3 (cost)	2026-05 ^b	usage logs	both tiers
Decontam. prompt control (App. J.4)	2026-06	question micro	gpt-4o
Session-granularity control (§4)	2026-06	question micro	gpt-4o
Tab. 8 (ablation)	Jan	conv. mean	gpt-4o
Tab. 14 (recall)	Jan	conv. mean	none
Tab. 15 (RAG grid)	2026-06	conv. macro	gpt-4o-mini
Tab. 18–19 (GraphRAG)	2026-06	question micro	gpt-4o-mini
Tab. 20 (thresholds)	Dec pilot	per-question	gpt-4o-mini
Fig. 2 (dial + anchors)	2026-06	question micro	gpt-4o ^c
App. F (SeCom dial)	2026-06	question micro	gpt-4o
App. G (sentence control)	2026-06	question micro	gpt-4o-mini
Tab. 7 (stateful probe)	2026-06	question micro	gpt-4o
App. N (retrievers)	2026-06	question micro	gpt-4o-mini
Tab. 17 (refusal)	2026-06	question micro	gpt-4o-mini
App. I (PerLTQA)	2026-06	conv. macro	gpt-4o-mini
Variance re-runs (§6)	2026-07	question micro	gpt-4o
App. H (EDU control)	2026-07	question micro	gpt-4o

Table 13: Provenance of every quantitative result.

^aGraphRAG row uses gpt-4o-mini (Table 1 footnote).

^bInstrumented cost runs of 2026-05-27; the chunks row was measured on the 2026-06 batch (Table 3 footnote). ^cDashed dial curve uses gpt-4o-mini (§6). ^dTwo-batch tables: the chunks rows (and, in Table 5, the RAG+Rerank row) are 2026-06; all other rows are 2025-12 under the identical protocol (table captions mark the affected comparisons). “Conv. mean/macro” rows are internally comparable but not commensurate with question-micro rows. Unless a caption states otherwise, Δ /gap values are computed from unrounded counts and may differ from the difference of the rounded cells by 0.1pp. Same-configuration repeats of the headline pipelines agree to ≤ 0.4 pp within a batch (three-run study, §6); the widest same-configuration repeat we observed is 2.0pp (SeCom at rate 0.75, Appendix F); cross-batch anchors drift 1–3.5pp with direction preserved and the drift is answerer-side (§6): chunks 43.9/45.2/47.4 (gpt-4o), 41.9/42.8/43.1 (gpt-4o-mini), RAG grid 55.5/52.5 (Table 15).

28.0%. Reproducing the live end-to-end pipeline additionally requires an OpenAI-compatible endpoint and the third-party benchmark files, which the README documents but which we do not redistribute.

J.3 Synthetic-Benchmark Metric Definitions

Let a denote a system’s answer, g the ground truth, and $\text{norm}(\cdot)$ lower-casing plus punctuation stripping. **Recall Rate** is the fraction of planted facts f with token-level fuzzy match $\text{FuzzyRatio}(a, f) \geq 0.80$ (via rapidfuzz); **Exact Match** is $\mathbb{K}[\text{norm}(g) \subseteq \text{norm}(a)]$ (substring containment). **Keyword Coverage** is $\text{KC}(a, g) = |K(g) \cap T(a)|/|K(g)|$ over curated keyword sets $K(g)$ and answer tokens $T(a)$; **Pass Rate** is $\mathbb{K}[\text{KC} \geq 0.80]$.

J.4 Prompt Templates

This section reproduces, verbatim, every prompt used in the experiments: the artifact-extraction prompt and its gleaning pass (write time), the answer-generation prompt (query time), and the LLM-judge prompts for both benchmarks. Long lines are re-wrapped to the column width and Unicode arrows are rendered as $\rightarrow$; two Chinese lexical cues in the gleaning prompt are transliterated in brackets. Content is otherwise character-for-character identical to the prompts in the released code.

Disclosure and decontamination control: eval-derived few-shot examples. The extraction prompt’s few-shot examples and geographic-entity rules were engineered on LoCoMo-style conversational content (the examples paraphrase LoCoMo subject matter, and one overlaps the content of a test question). To quantify the effect rather than merely disclose it, we re-ran the full artifact pipeline on LoCoMo categories 1–3 with a decontaminated prompt variant: every few-shot example in the extraction and gleaning prompts replaced by fictional workplace/travel personas with identical structure, artifact-type coverage, and date formats, under a fresh extraction cache namespace. The result is statistically indistinguishable from the original run: 28.5% vs. 28.0% overall (41 vs. 38 discordant pairs, McNemar $p=0.82$); the chunks gap persists at 15.4pp (43.9% vs. 28.5%, 151 vs. 43 discordant pairs, $p<10^{-14}$); and the artifact pipeline’s temporal lead over RAG+Rerank survives (29.6% vs. 16.5%, +13.1pp, against +14.7pp with the original prompt). Test-set-derived examples therefore inflate neither the headline gap nor the secondary temporal claim. Any residual prompt-engineering effect still benefits the *artifact* pipeline only—the chunk pipelines invoke no LLM at write time—so it biases the comparison *against* our headline finding. The decontaminated variant ships alongside the original prompt (selected by an environment flag) so both the published numbers and the control remain reproducible.

Artifact extraction prompt (extractor: gpt-4o-mini, temperature 0.1).

You are an expert at extracting structured cognitive objects from dialogue using systematic reasoning.

Given a conversation turn (user message + assistant response), extract any of these object types:

****Task-oriented types:****

1. ****decision****: A choice or decision made (e.g., "Let's use PostgreSQL", "I decided to pursue counseling")
2. ****todo****: Action items, tasks to do (e.g., "Need to implement auth", "Planning to go camping next month")
3. ****key_fact****: Important facts, numbers, constraints (e.g., "Budget is \$50k", "The event is in June 2023")
4. ****reminder****: Preferences, rules to remember (e.g., "User prefers TypeScript", "Always meditate in morning")
5. ****insight****: Conclusions, learnings (e.g., "The bottleneck is in the database", "Exercise helps my anxiety")

****Personal/Social types (IMPORTANT for conversations about people):****

6. ****person_attribute****: Personal traits, identity, status (e.g., "Caroline is a transgender woman", "Melanie is married with two kids", "John is single", "She moved from Sweden 4 years ago")
7. ****event****: Activities or occurrences WITH time (e.g., "Attended LGBTQ support group on 7 May 2023", "Ran a charity race last Sunday", "Painting a sunrise in 2022")
8. ****relationship****: Interpersonal connections (e.g., "Caroline and Melanie are close friends", "Known each other for 4 years")

****CHAIN-OF-THOUGHT EXTRACTION PROTOCOL:****
Before extracting, think step-by-step:

1. ****WHO****: Identify all people, entities, or subjects mentioned
2. ****WHEN****: Identify all temporal expressions (dates, times, sequences like "before", "after")
3. ****WHAT****: Identify facts, decisions, events, relationships
4. ****WHY****: Identify causal relationships (constraints that led to decisions)
5. ****CONNECTIONS****: Link related information (e.g., "Budget constraint" $\rightarrow$ "Choice decision")

****EXTRACTION RULES:****

- Extract ALL factual information about people (identity, status, activities, preferences)
- Extract ALL events with their time expressions
 - TEMPORAL REASONING IS CRITICAL
- Extract ALL relationships mentioned
- Each object should be self-contained and understandable without context
- ****CRITICAL****: Include "citation" field with EXACT quote from dialogue
- Do NOT skip personal information - it is important!
- Pay special attention to TEMPORAL SEQUENCES (before/after, yesterday/today, dates)

****GEOGRAPHIC ENTITY RULES (CRITICAL FOR RETRIEVAL):****

- For cities: ALWAYS include both country AND city (e.g., "France (Paris)", "Canada (Toronto)")
- For locations: Use format "Country (City)" even if only city is mentioned
- Examples:

- * "visited Paris" -> extract as "visited France (Paris)"
- * "went to Nuuk" -> extract as "went to Greenland (Nuuk)"
- * "lives in Toronto" -> extract as "lives in Canada (Toronto)"

- This explicit format helps with later retrieval of country/city information

****CRITICAL TEMPORAL RULES:****

- Preserve dates EXACTLY as written: "7 May 2023", "June 2023", "the sunday before 25 May 2023"
- DO NOT convert specific dates to relative expressions
- Include dates in both "content" and "time_expression" fields

Output JSON array:

```
[
  {
    "type": "decision|todo|key_fact|reminder|insight|person_attribute|event|relationship",
    "content": "The extracted information INCLUDING exact dates if mentioned",
    "citation": "EXACT verbatim quote from dialogue",
    "context": "Brief explanation of why extracted",
    "confidence": 0.0-1.0,
    "time_expression": "VERBATIM time expression or empty string"
  }
]
```

****Example 1 (Social conversation):****

User: "Caroline is a transgender woman who moved from Sweden 4 years ago. She went to the LGBTQ support group on 7 May 2023."
 Assistant: "That's wonderful that she found a supportive community!"

Output:

```
[
  {
    "type": "person_attribute", "content": "Caroline is a transgender woman", "citation": "Caroline is a transgender woman", "context": "Identity information", "confidence": 0.95, "time_expression": ""},
  {
    "type": "person_attribute", "content": "Caroline moved from Sweden 4 years ago", "citation": "moved from Sweden 4 years ago", "context": "Background information", "confidence": 0.9, "time_expression": "4 years ago"},
  {
    "type": "event", "content": "Caroline attended LGBTQ support group on 7 May 2023", "citation": "went to the LGBTQ support group on 7 May 2023", "context": "Activity with specific date", "confidence": 0.95, "time_expression": "7 May 2023"}
]
```

****Example 2 (Mixed conversation):****

User: "Melanie is married with two kids. She's planning to go camping in June 2023. We've been friends for 4 years."
 Assistant: "Sounds like a fun family trip!"

Output:

```
[
  {
    "type": "person_attribute", "content": "Melanie is married with two kids", "citation": "Melanie is married with two kids", "context": "Family status", "confidence": 0.95, "time_expression": ""},
  {
    "type": "event", "content": "Melanie planning camping trip in June 2023", "citation": "planning to go camping in June 2023", "context": "Future activity with date", "confidence": 0.9, "time_expression": "June 2023"},
  {
    "type": "relationship", "content": "User and Melanie have been friends for 4 years", "citation": "We've been friends for 4 years", "context": "Friendship duration", "confidence": 0.9, "time_expression": "4 years"}
]
```

Gleaning prompt (second extraction pass).

You are reviewing an extraction result for missed information.

****Previous extraction found these objects:****
 {previous_objects}

****Original dialogue:****
 {dialogue}

****Your task:**** Find ANY information that was MISSED in the first extraction. Focus on:

1. Pronoun references ("She", "He" -> Who specifically?)
2. Omitted subjects (Who is doing the action?)
3. Implicit causality ([yinwei "because"], [dao zhi "leads to"] -> What causes what?)
4. Time expressions (dates, [zhi hou "after"], "before")
5. Relationships (Who knows whom?)
6. Location details (Where?)

****CRITICAL FORMAT RULES:****

- Output a JSON array
- Each object MUST have these fields: type, content, citation, context, confidence, time_expression
- If nothing was missed, output EXACTLY: []

****Example output (if found missed info):****

```
[{
  "type": "person_attribute", "content": "Caroline moved from Sweden 4 years ago", "citation": "moved from Sweden 4 years ago", "context": "Background missed", "confidence": 0.9, "time_expression": "4 years ago"}]
```

****Example output (if nothing missed):****
 []

Now find any missed objects:

Answer-generation prompt (answerer: gpt-4o, temperature 0, max 200 tokens; identical for every pipeline). The {context} slot receives the retrieved items—verbatim chunks, artifacts, or baseline context—with their turn indices and session timestamps.

Based on the following memory context, answer the question.

Memory Context
{context}

Question
{question}

Instructions
1. Identify relevant facts from the context
2. Connect facts if needed for multi-hop reasoning
3. Synthesize a complete answer

Answer
Provide a concise, direct answer.

LoCoMo judge prompt (judge: gpt-4o-mini, temperature 0, max 10 tokens; categories 1–4).

You are an expert evaluator. Judge if the predicted answer is semantically correct compared to the ground truth.

Question
{question}

Ground Truth Answer
{ground_truth}

Predicted Answer
{answer}

Evaluation Criteria (Binary - strict)
- CORRECT: The predicted answer correctly answers the question and conveys the same core meaning as the ground truth. Synonyms, paraphrases, and equivalent formats (e.g., "March 15" vs "3/15") are acceptable.
- INCORRECT: The predicted answer is wrong, incomplete, irrelevant, contradicts the ground truth, or fails to answer the question.

Important
- Be strict: partial answers or answers with significant missing information should be marked INCORRECT
- The answer must address the actual question being asked

Response Format
Reply with ONLY one word: CORRECT or INCORRECT

LoCoMo abstention judge (category 5). Mirrors the official LongMemEval abstention template; the dataset's trap answer is deliberately withheld from the judge (showing a reference answer pulls the judge back into semantic-match habits and passes trap-asserting answers).

I will give you a question that cannot be answered from the conversation it was asked about, and a response from a model. The correct behavior is to identify that the asked information is not available. Please answer yes if the model identifies the question as unanswerable -- for example, it says the information is not mentioned, is incomplete, or cannot be determined. Answer no if the model asserts a specific answer as fact.

Question: {question}

Model Response: {answer}

Does the model correctly identify the question as unanswerable? Answer yes or no only.

LongMemEval judge prompts. Exact copies of the official templates in LongMemEval's `evaluate_qa.py` (Wu et al., 2025), dispatched by question type. For single-session-user, single-session-assistant, and multi-session:

I will give you a question, a correct answer, and a response from a model. Please answer yes if the response contains the correct answer. Otherwise, answer no. If the response is equivalent to the correct answer or contains all the intermediate steps to get the correct answer, you should also answer yes. If the response only contains a subset of the information required by the answer, answer no.

Question: {question}

Correct Answer: {answer}

Model Response: {response}

Is the model response correct? Answer yes or no only.

The temporal-reasoning template appends to the criteria above: *"In addition, do not penalize off-by-one errors for the number of days. If the question asks for the number of days/weeks/months, etc., and the model makes off-by-one errors (e.g., predicting 19 days when the answer is 18), the model's response is still correct."* The knowledge-update template replaces the subset clause with: *"If the response contains some previous information along with an updated answer, the response should be considered as correct as long as the updated answer is the required answer."* The single-session-preference template judges against a rubric: *"I will give you a question, a rubric for desired personalized response, and a response from a model. Please answer yes if the response satisfies the desired response. Otherwise, answer no. The model does not need to reflect all the points in the rubric. The response is correct as long as it recalls and utilizes the user's personal information correctly."* The abstention template is:

I will give you an unanswerable question, an explanation, and a response from a model. Please answer yes if the model correctly identifies the question as unanswerable. The model could say that the information is incomplete, or some other information is given but the asked information is not.

Question: {question}

Explanation: {answer}

Model Response: {response}

Does the model correctly identify the question as unanswerable? Answer yes or no only.

J.5 Hyperparameters

Models: GPT-4o-mini (extraction), bge-m3 (embeddings). **Retrieval:** final top-15 from a reranked pool of 30; context cap 5000 tokens. **Temperature:** 0.1 (extraction), 0.0 (generation).

J.6 Judge Validation: Human-Agreement Study

We validated the GPT-4o-mini judge against human annotation on a 100-question subset, stratified 25 per benchmark $\times$ representation cell (LoCoMo / LongMemEval-S $\times$ chunks / artifacts) and sampled from the same runs as Tables 1 and 2. The annotation sheet was blind: it contained only the (q, g, a) triple—the same information condition the judge receives—with judge verdicts and pipeline identity withheld. A human annotator labeled every item (LLM-assisted pre-screening with full manual verification; all flagged borderline items adjudicated by the annotator). Agreement is **95%** with Cohen’s $\kappa = 0.897$. Disagreements are near-symmetric—2 human-only-correct vs. 3 judge-only-correct—so the judge is neither systematically lenient nor strict, and agreement is balanced across the axes that matter for our comparison: chunks $\kappa=0.92$ vs. artifacts $\kappa=0.86$ (no representation favoritism), LoCoMo $\kappa=0.88$ vs. LongMemEval-S $\kappa=0.92$, and short answers (≤ 30 whitespace tokens) $\kappa=0.89$ vs. long answers $\kappa=1.00$ —ruling out the concern that the judge rewards the artifact pipeline’s terser answer style. All five disagreements are boundary cases (partial enumerations or granularity mismatches between gold and answer), not directional errors.

To address the concern that a single author-annotator could be biased, an *independent* annotator (not an author, with no stake in the outcome) relabeled the same 100 items under the identical blind protocol. Inter-annotator agreement (independent vs. author) is **90%** ($\kappa=0.80$, substantial), and independent-vs-judge agreement is **89%** ($\kappa=0.78$). The independent annotator’s disagreements with the judge are one-directional—10 of 11 are answers the human credits but the judge does not—so this annotator is slightly more lenient than the judge, chiefly on temporal precision (accepting “2022” for “November 7, 2022”) and on preference-match questions. Because such leniency credits *more* answers correct and falls disproportionately on chunks (chunks agreement 86% vs. artifacts 92%), it can only widen the measured chunk–artifact gap on this full-distribution sample, leaving the head-

line gap conservative rather than inflated.

The headline ordering is also metric-independent: recomputing LoCoMo’s official lexical token-F1 (normalization, Porter stemming, token-level F1) offline from the stored answer strings of the two headline runs gives chunks 31.1 vs. artifacts 26.1 ($n=699$ each), with chunks ahead in every category—the same ordering as the LLM judge, compressed in magnitude because lexical F1 awards partial credit to incomplete answers.

The ordering is judge-*family*-independent as well. To rule out that the gap is an artifact of the GPT-4o-mini judge specifically—e.g. an LLM judge over-crediting the verbatim pipeline’s lexical overlap with the gold—we re-scored the *stored answers* of both headline runs with a panel of seven judges spanning five model families from independent vendors: OpenAI (gpt-4o-mini, gpt-4o), Alibaba (qwen-plus, qwen-max), Google (gemini-2.5-flash-lite), Anthropic (claude-haiku-4.5), and DeepSeek (deepseek-v3.2), under the identical per-type prompts and binary protocol (no retrieval or generation re-run). Every judge preserves the chunk–artifact gap, and the paired McNemar exact test (the headline test, recomputed per judge) is significant for all fourteen benchmark $\times$ judge cells: on LoCoMo the gap ranges +13.0 to +17.0pp ($p < 10^{-10}$ throughout; e.g. the cross-vendor claude-haiku-4.5 41.6 vs. 28.2, discordant 142/48, $p=5.5 \times 10^{-12}$), and on LongMemEval-S +19.4 to +23.8pp ($p < 10^{-12}$ throughout; e.g. deepseek-v3.2 65.8 vs. 44.4, discordant 152/45, $p=9.0 \times 10^{-15}$). The four non-OpenAI vendors share no training lineage with the GPT judge of the main tables yet reproduce the full effect, and re-judging with GPT-4o-mini itself recovers the published headline within 0.6pp (44.2/27.9 and 68.0/46.0). The comparison is thus stable across judge families, not only across the lexical-vs-LLM metric boundary.

The category-5 abstention judge (a LongMemEval-style abstention template) is *not* covered by the human study above, which sampled answerable questions only. As an offline consistency check, we compared its verdicts against an independent lexical abstention detector (a regex over refusal phrasings) on all 446 category-5 answers per pipeline: raw agreement is 97.5% on the chunks run and 92.4% on the artifacts run ($\kappa=0.76/0.63$; the modest κ values reflect low abstention base rates of 6.5%/15.0%). Disagreements are one-sided—in 42 of 45 cases

the LLM judge credits a paraphrased refusal that the regex misses—so the lexical check behaves as a stricter subset of the judge, not a contradicting signal. As a second check, an independent LLM annotator—run in a fresh context, blind to pipeline identity, judge verdicts, and all paper statistics—labeled a random sample of 50 category-5 answers (25 per pipeline, fixed seed) under the same abstention criterion: it reproduced the judge’s verdict on all 50 (5 abstentions, 45 answers; $\kappa=1.0$), with no items flagged as borderline. Finally, the *independent human* annotator above labeled the same 50 category-5 answers under the abstention criterion, blind to judge verdicts: agreement with the judge is **100%** ($\kappa=1.0$; 5 abstentions, 45 answers), the first direct human validation of the abstention judge.

J.7 Fair-Comparison Disclosure: Per-Method Tuning Effort

To preempt the concern that our cross-method comparison rests on an under-tuned set of baselines, we document the hyperparameter-search effort invested in each retrieval-augmented method. For **RAG** we use common community defaults—512-character chunks, 100-character overlap, dense top- $k=10$ —which every chunk-based pipeline in the paper, including the verbatim-chunk variant, inherits unchanged. A chunk-size $\times$ top- k grid re-run under the main synthetic protocol (Appendix M) confirms the inherited configuration is not a cherry-picked optimum—larger chunks score higher on that probe—and any residual under-tuning of RAG biases *against* our headline finding, since RAG sits on the verbatim side of the comparison and a stronger RAG could only widen the artifact pipeline’s deficit. For **RAG+Rerank** the reranker stage adds a coarse pool size (fixed at top-20 from the dense retriever, $2\times$ the final k) and a final top- k (10, matching vanilla RAG); chunking is identical to RAG. For **GraphRAG** we ran a two-tier 4-axis grid sweep over chunk_size, max_gleanings, community_level, and search_method—9 unique configurations across 12 runs in total (Appendix O)—and report the best cell. For **summarization** the GPT-4o-mini summarizer was prompt-engineered on a held-out set of 5 conversations before evaluation. **CogCanvas** hyperparameters were fixed by ablation on the synthetic benchmark and held constant across LoCoMo and LongMemEval; we did not retune them per-dataset, with one disclosed exception: the extraction prompt’s few-shot examples

target LoCoMo-style conversational content (Appendix J.4), an effort that benefits only the artifact pipeline and therefore makes the headline comparison conservative; a decontaminated-prompt re-run confirms the numbers are unchanged (28.5% vs. 28.0%, McNemar $p=0.82$). Each numeric entry in Table 1 thus reflects either a documented tuning effort (GraphRAG, summarization, CogCanvas) or deliberately standard settings whose under-tuning could only work against our conclusion (RAG, RAG+Rerank).

Threshold Selection. We tuned similarity thresholds on a held-out development set of 5 conversations (disjoint from evaluation data):

- θ_{ref} : Grid search over $\{0.3, 0.5, 0.7\}$ selected $\theta_{\text{ref}} = 0.5$ as optimal for balancing precision and recall in reference edge creation.
- θ_{causal} : Set to $\theta_{\text{ref}} - 0.05 = 0.45$ to allow slightly looser causal connections while maintaining semantic coherence.

Sensitivity analysis (Appendix P) confirms robustness: performance stays within a 10pp band (57.5%–67.5%) across the tested configurations, indicating the method is not overly sensitive to threshold selection.

K Retrieval Quality Analysis

Config.	Overall	M-hop	Temp.	O-dom.
Full System	71.8%	87.3%	63.2%	55.0%
– Graph Exp.	25.8%	32.3%	20.5%	24.7%
Improvement	+45.9pp	+55.0pp	+42.6pp	+30.3pp
Chunk store*	59.0%	56.1%	67.6%	38.7%
+ rerank*	61.6%	61.2%	68.9%	38.4%

Table 14: Retrieval recall: percentage of ground-truth keywords found in retrieved context (no LLM answering; artifact rows January batch, same protocol as Table 8). Graph expansion improves the artifact store’s keyword recall by +46pp. *Chunk-store rows (no graph; second row after reranking): June batch, identical script and k .

Keyword recall is not a proxy for answer quality. The same probe applied to the verbatim-chunk store yields 59.0% before and 61.6% after reranking—ten points *below* the graph-expanded artifact store’s 71.8%, even though the chunk pipeline’s final answer accuracy is 15.9 points *higher* (Table 1). Keyword co-occurrence over-credits the artifact store: graph expansion retrieves many short artifacts whose union covers the gold

keywords without any single item carrying the answer-bearing wording. Two further observations: reranking shifts chunk-store recall by only +2.6pp, so the reranker’s training distribution (passage-length text) is not what manufactures the chunk advantage; and the store-level survival probe of Appendix D, which asks whether the gold survives in any *single* stored item, measures the informational question this keyword metric cannot.

L Cost Analysis Details

Extraction is not the dominant cost. Across 2,938 indexed turns, CogCanvas spent \$0.14 on per-turn extraction (gpt-4o-mini) against \$2.78 on answer generation (gpt-4o)—4.7% of total cost, amortizing to \$0.000047/turn: the answer-model call dominates regardless of memory architecture, contrary to the common concern that per-turn extraction is inherently expensive.

Extraction buys cheaper generation—and still loses on cost-effectiveness. The artifact pipeline’s gen cost (\$2.78) is 20.5% below RAG+Rerank (\$3.50) and 28% below chunks (\$3.84): artifacts pack a fact into 30–60 tokens versus ~ 130 per chunk, and answers from pre-structured artifacts are direct—completion tokens $5.1\times$ smaller (16.7 vs. 85.9 against RAG+Rerank), leveraged by gpt-4o’s $4\times$ output pricing. But the \$0.92 saved over chunks across 699 questions buys a 15.9-point accuracy drop: the chunks pipeline is simultaneously the most expensive per query and the cheapest per correct answer.

Deployment at a cheaper answerer scales linearly. Swapping the answerer to gpt-4o-mini drops the artifact pipeline from \$2.92 to \$0.31 ($9.5\times$), RAG/RAG+Rerank from \$3.50 to \$0.23 ($15.3\times$), and chunks from \$3.84 to \$0.23 ($16.7\times$; no extraction tail and the longest prompts). Re-pricing the original run’s tokens at mini rates agrees with the actual end-to-end re-run within $\pm 1.4\%$ for CogCanvas ($\pm 17\%$ for Summarization, whose summarize calls share ANSWER_MODEL), so Table 3 can be read at either tier.

GraphRAG costs are not instrumented. GraphRAG’s LLM calls run inside the Microsoft graphrag subprocess, bypassing our logger; published estimates (Edge et al., 2024) suggest $\sim \$35$ – 60 for indexing plus \$3–5 per query—an order of magnitude above CogCanvas.

Chunk size	Overlap	$k=5$	$k=10$
256	50	43.5%	50.5%
512	100	49.0%	52.5%
1024	200	63.0%	63.0%

Table 15: RAG pass rate on the synthetic multi-hop probe across chunk size and retrieval depth (same protocol and answerer as Table 5). The configuration used throughout the paper (512 chars, $k=10$) is *not* the grid optimum.

Latency. Per-query latency (compute-side) for the local BGE backends on LoCoMo10:

- **Embedding** (bge-m3): ~ 0.84 ms/query on average across CogCanvas, RAG, and RAG+Rerank.
- **Reranking** (bge-reranker-v2-m3): ~ 1.56 ms/query on average for CogCanvas and RAG+Rerank.

End-to-end wall-clock latency is dominated by the gpt-4o answerer call (~ 2.5 s/query for CogCanvas, ~ 2.5 s for RAG+Rerank). The headline cost figures (Table 3) and the gpt-4o-mini what-if are derived from full response .usage traces and the OpenAI list-price snapshot of 2026-05-27.

M RAG Retrieval Sensitivity

Every chunk-based pipeline in the paper inherits one retrieval configuration (512-character chunks, 100-character overlap, dense top- $k=10$). To check that this choice does not under-sell verbatim retrieval, we ran a chunk-size $\times$ top- k grid on the synthetic multi-hop probe under the protocol and answerer of Table 5 (50 conversations, rolling compression every 40 turns, gpt-4o-mini answerer). Table 15 reports pass rates; the inherited configuration re-measures at 52.5% against 55.5% in the main batch, a re-run difference within the cross-batch drift documented in Table 13 ($n=176$ questions).

Two observations. First, the inherited configuration is not a cherry-picked optimum: 1024-character chunks reach 63.0%, 10.5pp above the reported cell—and still 18.0pp below the artifact pipeline’s 81.0% on this home-turf probe, so no cell changes any qualitative conclusion. Because RAG and the chunk pipelines sit on the *verbatim* side of the central comparison, any residual under-tuning makes the headline finding conservative. Second, the ranking ($1024 > 512 > 256$; $k=10 \gtrsim k=5$) replicates under a gpt-4o answerer (absolute range

Retriever family	Chunks	Artifacts	Gap
bge-m3 (dense, default)	43.1	28.5	+14.6
text-embedding-3-small (dense)	43.5	29.9	+13.6
BM25 (sparse lexical)	43.8	29.0	+14.7

Table 16: LoCoMo categories 1–3 (699 questions, LLM-Judge %), chunks vs. artifacts under three retriever families, with the answerer fixed at gpt-4o-mini and extraction recomputed without cache. The chunk advantage is stable at +13.6 to +14.7pp; chunks barely move (43.1–43.8) as the retriever changes.

47.0–70.5%), so it is not an answerer artifact. We nevertheless keep the community-standard 512-character configuration everywhere: the probe is the one setting we explicitly flag as a fragile basis for architectural claims (§4), and tuning the shared chunking on it would contradict the no-per-dataset-tuning policy of Appendix J.7.

N Retriever-Family Robustness and Abstention Repair

The ordering survives a retriever-family swap.

The main comparison fixes one retriever (bge-m3 embedder, bge-reranker-v2-m3). To test whether the chunks-over-artifacts ordering is specific to that family, we re-ran the LoCoMo categories 1–3 comparison (699 questions, gpt-4o-mini answerer held fixed, extraction recomputed from source with the cache disabled so that every store is re-embedded under the retriever actually in use) under three retriever families: the default dense bge-m3, a held-out dense embedder (OpenAI text-embedding-3-small, 1536-dim), and a sparse lexical retriever (BM25, no embeddings at all). Table 16 shows the chunk advantage is essentially invariant: chunks score 43.1–43.8% and artifacts 28.5–29.9% regardless of family, leaving a gap of +13.6 to +14.7pp. The negative result is therefore not an artifact of the bge-m3 embedder, and it persists even when retrieval is switched from dense semantic matching to sparse lexical overlap—the regime that, if anything, should most favor a retriever’s ability to surface raw source text.

Chunk size on LoCoMo. Complementing the synthetic-probe chunk-size grid of Appendix M, on LoCoMo itself enlarging the verbatim window from 512 to 768 characters (overlap 100→150) lifts chunks from 43.1% to 47.2% on categories 1–3 (gpt-4o-mini). The headline 512-character setting is thus conservative for the verbatim side, and the

ordering does not depend on chunk-size tuning.

Three refusal mechanisms confirm the abstention trade-off is intrinsic.

We tested whether chunks’ abstention weakness (§6) admits repair, escalating from a cheap score gate to two stronger semantic refusers (all gpt-4o-mini). The *score-threshold gate* refuses when the top reranker score falls below τ (and the prompt rule discouraging refusal is removed); sweeping τ (Table 17, top) lifts category-5 refusal from the ungated 6.5% to 55.6–65.5%, approaching naive RAG’s competence, but answerable accuracy on categories 1–3 collapses from 43.1% to 25.3–17.9%—below even the artifact pipeline (28.5%). Because a raw similarity score cannot separate a topically-relevant-but-unanswerable retrieval from an answerable one, we added two mechanisms with a *semantic* signal: (M1) an *LLM evidence-sufficiency gate* that asks the answerer model whether the retrieved verbatim evidence actually contains the answer before responding, and (M2) *answer-then-verify*, which answers normally and then asks whether the answer is supported by the retrieved evidence, overriding to a refusal if not. Neither escapes the frontier (Table 17, bottom; matched gpt-4o-mini no-gate 44.6/10.8): M1 only modestly Pareto-improves on the score gate—+3.7pp answerable at a matched ~59% abstention (25.6% vs. 21.9%)—and M2 lands on it (22.3%/58.1%). Reaching ~59% category-5 refusal still costs ~19pp answerable accuracy (44.6→25.6%; McNemar 146 answerable questions newly refused vs. 13 recovered, $p < 10^{-24}$). The pattern **replicates on LongMemEval-S** (M1 vs. matched no-gate, gpt-4o-mini): abstention rises 53.3→93.3%—past naive RAG’s 70%—but answerable accuracy falls 63.2→50.2% (McNemar 65 lost vs. 4 gained, $p < 10^{-12}$). A similarity gate, an LLM sufficiency judge, and an answer verifier thus all fail to install competent refusal without sacrificing answerable questions; that the refusal rate *can* be driven past naive RAG, just not for free, points to the same remedy as our main finding—a refusal/verification layer should *augment* the verbatim store rather than be bolted onto retrieval. Competent refusal is a genuine property the verbatim representation lacks, not a tuning artifact.

O GraphRAG Baseline Sensitivity

To address reviewer concerns that the GraphRAG baseline in Table 1 might be under-tuned, we con-

Refusal mechanism	Cat-5 abstention	Cat 1–3 answerable	Configuration	Overall	M-hop	Temp.	O-dom.
<i>Score-threshold gate</i> (no gate: 6.5 / 43.1)			cs=800, g=3, cl=2, local	13.5	16.2	2.2	52.4
$\tau=0.1$	55.6	25.3	cs=800, g=1, cl=2, local (default)	13.0	18.9	2.2	38.1
$\tau=0.2$	59.2	21.9	cs=800, g=2, cl=2, local	12.4	16.2	2.2	42.9
$\tau=0.3$	63.5	17.9	cs=800, g=1, cl=3, local	12.4	16.2	2.2	42.9
$\tau=0.4$	65.5	—	cs=800, g=1, cl=2, drift	11.9	13.5	4.4	38.1
			cs=400, g=1, cl=2, local	11.9	16.2	1.1	42.9
			cs=1200, g=1, cl=2, local	11.4	13.5	2.2	42.9
<i>Semantic refusers</i> (no gate: 10.8 / 44.6)			cs=800, g=1, cl=1, local	10.8	13.5	2.2	38.1
M1 sufficiency, $\tau=0.0$	59.4	25.6	cs=800, g=1, cl=2, global	5.4	6.8	0.0	23.8
M1 sufficiency, $\tau=0.1$	62.1	19.7					
M1 sufficiency, $\tau=0.2$	64.8	17.2					
M2 answer-then-verify	58.1	22.3					

Table 17: Refusal mechanisms on the chunk pipeline (LoCoMo, gpt-4o-mini, LLM-Judge %). *Top*: an evidence-sufficiency gate on the top reranker score; as τ rises, category-5 refusal climbs toward naive-RAG levels but answerable accuracy (categories 1–3) falls monotonically below even the artifact pipeline (the $\tau=0.4$ answerable cell is omitted, abstention already saturating). *Bottom*: two stronger *semantic* refusers—an LLM evidence-sufficiency gate (M1, swept over a coarse score pre-filter τ) and answer-then-verify (M2). M1 modestly Pareto-dominates the score gate at matched abstention, but no mechanism escapes the trade-off. The two blocks use separate matched gpt-4o-mini no-gate anchors (6.5/43.1 and 10.8/44.6, equal within run-to-run noise).

ducted a 4-axis grid search over the Microsoft graphrag library’s principal configuration knobs and report the per-cell accuracy on LoCoMo. *Aggregation note*: all GraphRAG numbers in this paper—the grid tables below and the main results table—are per-question micro-averages, matching the main-table protocol. (The sweep harness additionally logs per-conversation macro-averages, e.g. 12.7% for the drift winner; cell rankings are identical under either aggregation.)

Search space. Four axes: (i) **chunk_size** (index-time) $\in \{400, 800, 1200\}$; (ii) **max_gleanings** (index-time, number of additional entity-extraction passes per chunk) $\in \{1, 2, 3\}$; (iii) **community_level** (query-time, depth in the hierarchical community summary tree) $\in \{1, 2, 3\}$; (iv) **search_method** (query-time) $\in \{\text{local}, \text{global}, \text{drift}\}$.¹ The full Cartesian product spans 81 configurations.

Two-tier protocol. To make the sweep tractable, we ran a two-tier procedure. **Tier 1** enumerated

¹The drift mode is the library’s dynamic retrieval-informed token-level filtering method introduced in graphrag 2.x; it most closely approximates a “hybrid” search semantically. We exclude basic mode because it falls back to plain vector retrieval and does not exercise the GraphRAG pipeline.

Table 18: Tier 1 grid search on a 3-conversation LoCoMo subset, 185 questions (LLM Judge, %). The default configuration ranks second; the top row is the max_gleanings = 3 leader. The open-domain column shows substantial variance across cells (23.8–52.4) reflecting the small denominator (21 open-domain questions on the subset).

the default configuration plus all 1-D excursions ($1 + 4 \times 2 = 9$ configurations) on a 3-conversation subset of LoCoMo (locomo_000, _001, _002), identifying which axes carry signal. **Tier 2** ran the three Tier-1 leaders on the full 10-conversation LoCoMo benchmark (699 questions, categories 1–3, LLM Judge evaluation) to produce the final tuned-baseline numbers. Total Tier 1 wall time was 7.1 h; Tier 2 was 10.5 h, dominated by the slower drift mode (519 min for one cell).

Tier 1 (3 conversations, 9 cells). Table 18 reports the 1-D sweep. Three findings are robust: (1) chunk_size and community_level have only modest effects (−0.6 to −2.2pp overall vs. the default); (2) max_gleanings = 3 is the strongest index-time signal, lifting open-domain by 14.3pp on the 3-conversation subset; (3) global mode collapses to 5.4% overall and should not be used for conversational memory. Crucially, the temporal-accuracy column stays at or below 2.2% in 8 of 9 cells—only drift mode moves the needle, and even then only to 4.4% (the full-benchmark comparison follows in Tier 2).

Tier 2 (full LoCoMo, 3 cells). The Tier-1 leaders—max_gleanings = 3 (overall + open-domain winner), the default (multi-hop winner), and drift mode (sole temporal mover)—were re-run on the full 10-conversation LoCoMo (Table 19). Two findings: (1) the open-domain advantage of max_gleanings = 3 disappears at scale, regressing to 31.2%—a known small-sample artifact of the 3-conversation subset; (2) drift mode emerges as the across-the-board winner (overall 13.0%, temporal 7.5%, open-domain 32.3%), reversing its mid-pack Tier 1 ranking. We therefore report the drift

Configuration	Overall	M-hop	Temp.	O-dom.
cs=800, g=1, cl=2, drift (tuned)	13.0	12.8	7.5	32.3
cs=800, g=3, cl=2, local	11.4	13.8	3.4	31.2
cs=800, g=1, cl=2, local (default)	11.2	13.5	3.1	31.2
Verbatim chunks, no graph*	43.9	35.5	50.2	47.9
Artifacts + graph*	28.0	22.3	31.2	34.4
Gap: tuned GraphRAG – chunks	−30.9	−22.7	−42.7	−15.6

Table 19: Tier 2 grid search on the full LoCoMo benchmark (10 conversations, 699 questions, LLM Judge %). The drift cell is the tuned ceiling reported in the main results table. *Reference rows reproduce the main-table numbers (Table 1); all rows are per-question micro-averages (see aggregation note above).

cell as the GraphRAG row in Table 1.

Interpretation. Even after two-tier tuning across four axes (9 unique configurations, 12 runs) spanning index-time and query-time hyperparameters, GraphRAG’s temporal accuracy on LoCoMo remains capped at 7.5%—a 42.7pp deficit relative to verbatim chunks, and 23.7pp below even the extracted-artifact pipeline. The narrowness of this band across 9 distinct configurations indicates that the limitation is not configurational but *paradigmatic*: GraphRAG’s community-level summarization pipeline systematically abstracts away the verbatim timestamps that LoCoMo’s temporal questions demand. This is precisely the failure mode illustrated by the May 7, 2023 case study in §E—the entity “LGBTQ support group” survives summarization, but the date does not. Hyperparameters cannot recover information the indexing layer never preserved.

P Threshold Sensitivity

Table 20 shows CogCanvas is robust to threshold selection: pass rate on the synthetic multi-hop probe ranges only 10pp (57.5%–67.5%) across configurations. The sweep is a December pilot (10-conversation subset, single compression at turn 40 rather than the main rolling protocol, gpt-4o-mini answerer): its rows are internally comparable, but the absolute level is not commensurate with the 81.0% of Table 5.

Q Extended Related Work

This section provides detailed discussion of related work summarized in Section 2.

Config	Ref.	Causal	Pass
Low	0.3	0.25	62.5%
Default	0.5	0.45	67.5%
High	0.7	0.6	60.0%
Very High	0.8	0.7	57.5%

Table 20: Threshold sensitivity on the synthetic multi-hop probe (10-conversation pilot, pre-rolling protocol, gpt-4o-mini answerer; rows internally comparable only). Performance stable across values.

Q.1 Context Window Management

The finite context window of transformer-based LLMs has motivated extensive research into efficient context utilization. Sparse attention mechanisms (Child et al., 2019; Beltagy et al., 2020) reduce the quadratic complexity of self-attention, enabling longer sequences but not fundamentally solving the capacity constraint. Retrieval-augmented approaches (Borgeaud et al., 2022; Izacard et al., 2022) maintain external datastores that are queried at inference time, yet these systems typically assume static document collections rather than dynamically evolving conversational state. Recent work on context extension through positional interpolation (Chen et al., 2023) pushes the practical context limit but does not eliminate the fundamental trade-off between context length and computational cost. Our approach is orthogonal to these efforts: rather than extending raw context capacity, we focus on *what* to preserve when compression becomes necessary.

Q.2 Memory-Augmented Language Models

MemGPT (Packer et al., 2023) pioneered a hierarchical memory system inspired by operating system design, introducing virtual context management where LLMs perform explicit memory management operations (moving data between “fast” and “slow” memory tiers). The original MemGPT project has since evolved into Letta,² a production-ready framework that extends the core architecture with filesystem-based memory storage, PostgreSQL persistence, and enhanced tool integration. Letta’s recent LoCoMo evaluation achieved 74.0% accuracy using GPT-4o mini (Letta AI, 2024) by storing conversation histories as files with semantic search capabilities. While this demonstrates the power of stateful, database-backed memory systems, it requires persistent infrastructure (PostgreSQL, Docker) that introduces deployment

²<https://letta.ai>

complexity. This architectural approach represents a fundamentally different paradigm from our stateless, inference-time retrieval.

Several concurrent approaches explore alternative memory architectures. MemoryBank (Zhong et al., 2024) incorporates a memory updating mechanism inspired by the Ebbinghaus Forgetting Curve, allowing selective memory preservation based on temporal decay and significance. Long-Mem (Wang et al., 2023) employs a decoupled architecture with a frozen backbone LLM as memory encoder and a side-network for retrieval, caching attention key-value pairs from previous segments. ReadAgent (Lee et al., 2024) takes a human-inspired approach, compressing long documents into “gist memories” while retaining the ability to look up original passages when details are needed. These methods demonstrate creative solutions to the memory challenge, though they typically require either architectural modifications or focus primarily on document comprehension rather than dynamic conversational state.

More recently, several approaches have pushed the boundaries of memory-augmented LLMs through dedicated optimization. Most notably, EverMemOS (EverMind AI, 2025) achieves a breakthrough 92.3% on LoCoMo through categorical memory extraction (situational context, semantics, user profiling) and MemCell atomic storage with rich metadata (timestamps, source, tags, relational links). EverMemOS represents the current state-of-the-art, demonstrating that structured memory with explicit forgetting mechanisms can even surpass full-context approaches.

In contrast, the concurrent EMem (Zhou and Han, 2025) reaches 78.0% on LoCoMo (GPT-4o-mini answerer) while remaining entirely training-free: it decomposes each session into enriched Elementary Discourse Units—short, near-verbatim event propositions carrying normalized entities and source-turn attributions—and retrieves over them, explicitly aiming to preserve information “in a non-compressive form.” Its strong result with a representation that sits near the verbatim end of the fidelity spectrum is the outcome our account in §4.7 predicts, and we treat it as the near-verbatim data point in that reconciliation rather than as a competing system claim.

Two further neighbors deserve explicit positioning. Zeng et al. (2024) is the closest prior comparison: holding the model and embedder fixed, it sweeps chunks, knowledge triples, atomic facts,

summaries, and mixed stores across four tasks and six datasets (including LoCoMo), finding that different structures suit different tasks, that chunk-based stores excel on lengthy-context dialogue understanding, and that mixed stores are the most resilient. Our results agree with that ordering where the studies overlap; what we add is the controlled isolation of the *extraction-vs-source-text* premise on long-conversation benchmarks—paired McNemar statistics over a single fixed retrieval–rerank–reasoning pipeline, six confound controls, mechanism probes (store-level survival, keyword-recall audit), decontaminated-prompt and session-granularity reruns, and per-correct-answer cost accounting—together with the negative-result framing that the replacement premise itself fails. Zep (Rasmussen et al., 2025) reports strong Long-MemEval results with the Graphiti temporal knowledge graph; its lowest tier is an *episode sub-graph* that retains the raw interaction data beneath the entity and community layers, so it instantiates structure-annotating-text rather than structure-replacing-text and is therefore consistent with, not a counterexample to, the fidelity ordering of §4.7 (its absolute numbers come from a different answerer and harness and are not commensurable with our tables).

Heavily-optimized systems such as EverMemOS occupy a fundamentally different design space: they require substantial training infrastructure, domain-specific corpora, dedicated fine-tuning pipelines, or complex architectural modifications. While ideal for high-stakes deployments where maximum accuracy justifies the engineering investment, they are less suitable for practitioners who need immediate solutions without fine-tuning overhead.

Our study occupies the *training-free, stateless* corner of this design space deliberately: it is the setting where the stored representation can be swapped in isolation, without fine-tuning or infrastructure confounds. We do not compare absolute numbers against systems evaluated on different answerer models and evaluation stacks (EverMemOS, EMem); our contribution is the controlled within-pipeline comparison, which finds that the replacement premise common to lossy-extraction designs does not survive isolation—verbatim chunks dominate extracted artifacts, and the artifact pipeline does not beat naive RAG (Tables 1 and 2).

Q.3 Graph-Based Knowledge Organization

GraphRAG (Edge et al., 2024) and related approaches (Yasunaga et al., 2021; Zhang et al., 2022) leverage graph structures to organize and retrieve knowledge. These systems excel at capturing relational information and enabling multi-hop reasoning over structured knowledge bases. However, existing graph-based methods are predominantly designed for *static document collections*—corpora that are processed offline and queried at inference time. The dynamic, incremental nature of conversational knowledge poses distinct challenges: stored items must be produced in real time, any graph structure must evolve with each turn, and retrieval must balance recency with relevance. The artifact pipeline under test meets these requirements through online extraction and incremental graph construction—and our ablation shows that meeting them does not rescue the representation: the same incremental machinery over verbatim chunks performs better at every question type (Table 1).

Q.4 Conversation Summarization

Summarization-based approaches (Tang et al., 2021; Chen et al., 2021) offer an intuitive solution to context overflow: compress older turns into concise summaries that preserve essential information within reduced token budgets. Recent work has explored hierarchical summarization (Wu et al., 2021), where summaries are themselves summarized as conversations extend. While effective for capturing high-level narrative flow, summarization fundamentally operates on a lossy compression principle. Our experiments confirm that even state-of-the-art LLM-based summarizers systematically lose specific details; typed artifact extraction, despite its verbatim quote fields, only *mitigates* this loss (§4.7).